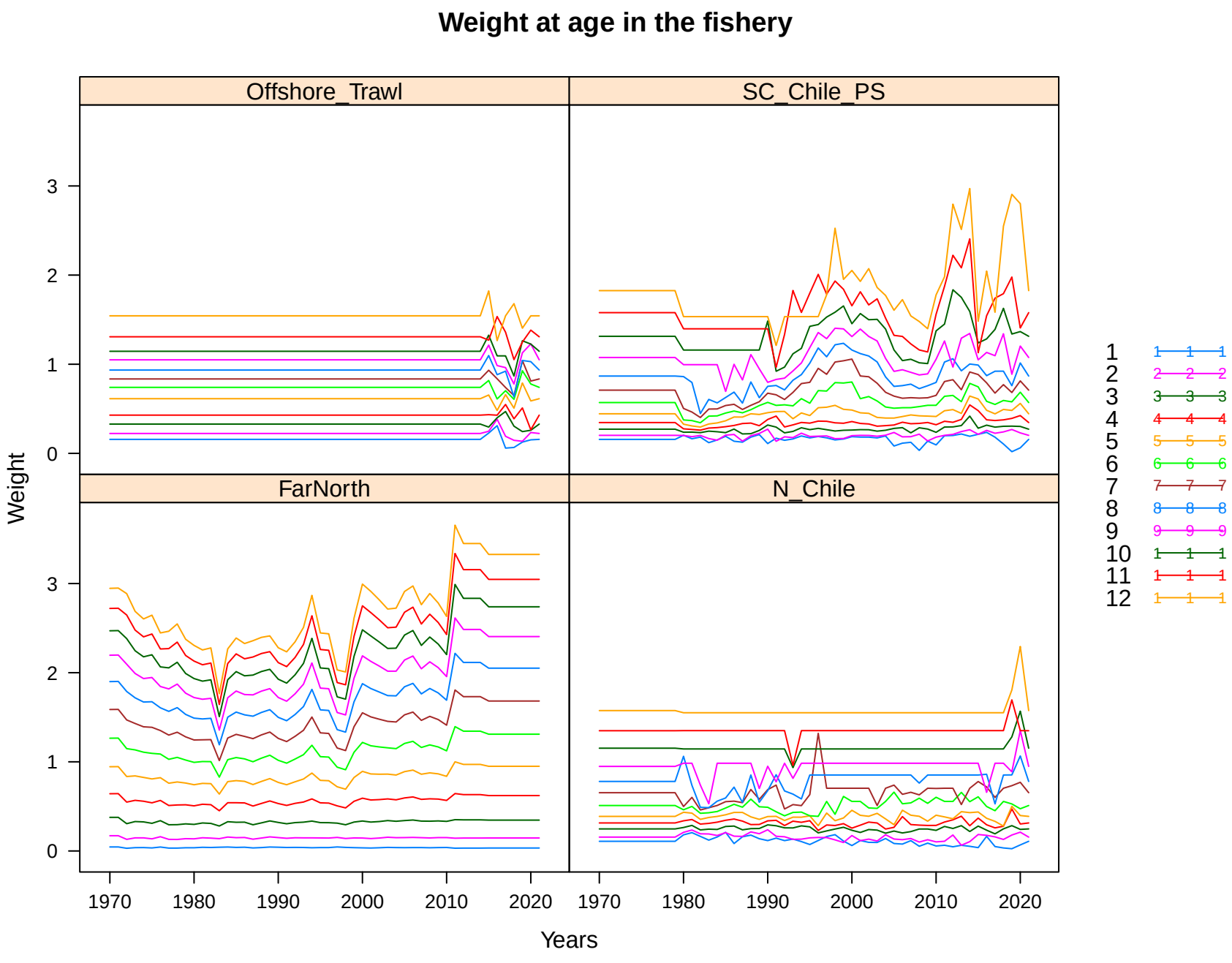
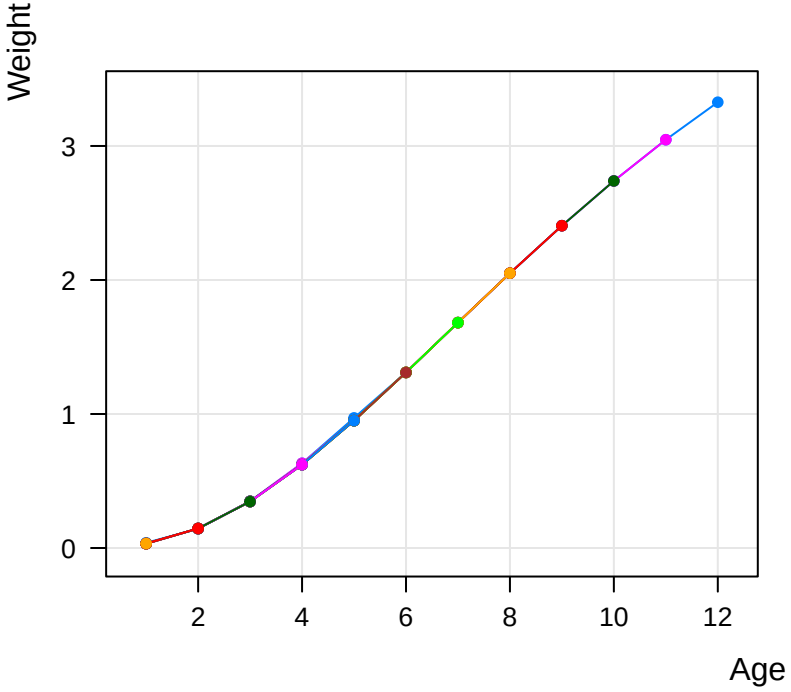
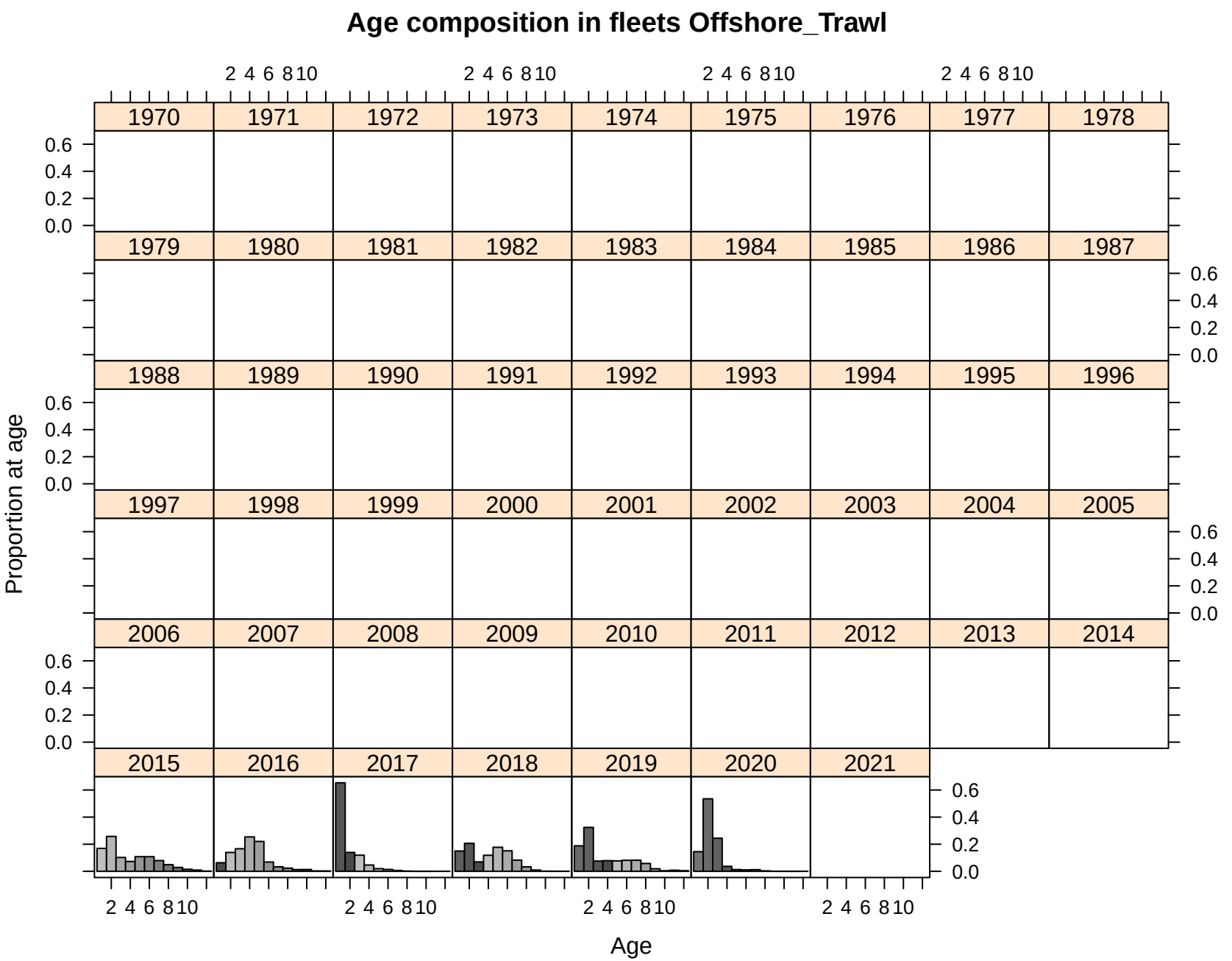

Weight at age in the fishery
Offshore_Trawl
SC_Chile_PS
FarNorth
N_Chile
Weight
Years
0
1
2
3
1970
1980
1990
2000
2010
2020
1
2
3
4
5
6
7
8
9
10
11
12

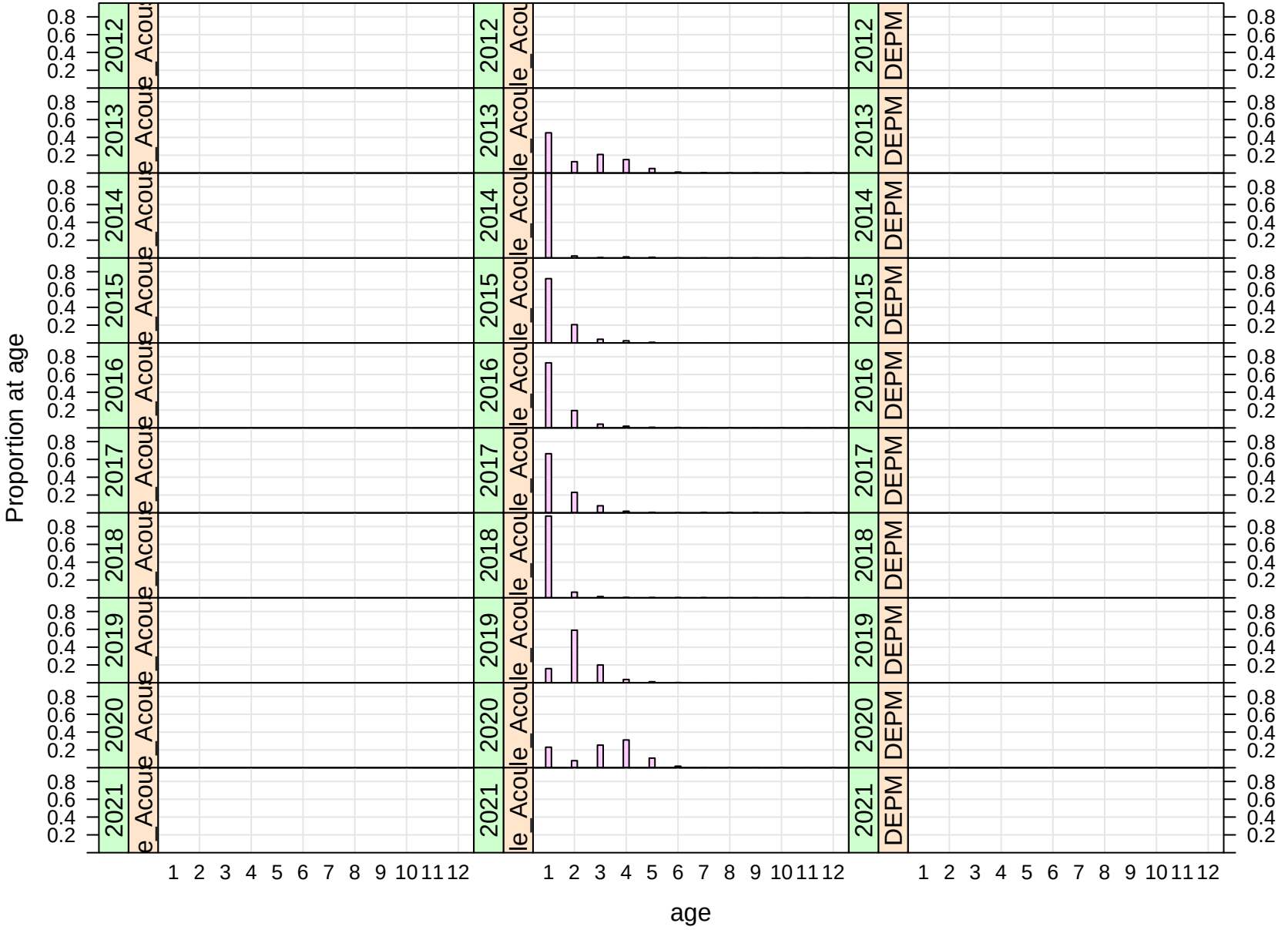
Weight at age in the survey
Offshore_CPUE
Peru_Acoustic
Peru_CPUE
Chile_AcousCS
Chile_AcousN
Chile_CPUE
DEPM
Weight
Years
0
1
2
3
1970
1990
2010
1
2
3
4
5
6
7
8
9
10
11
12

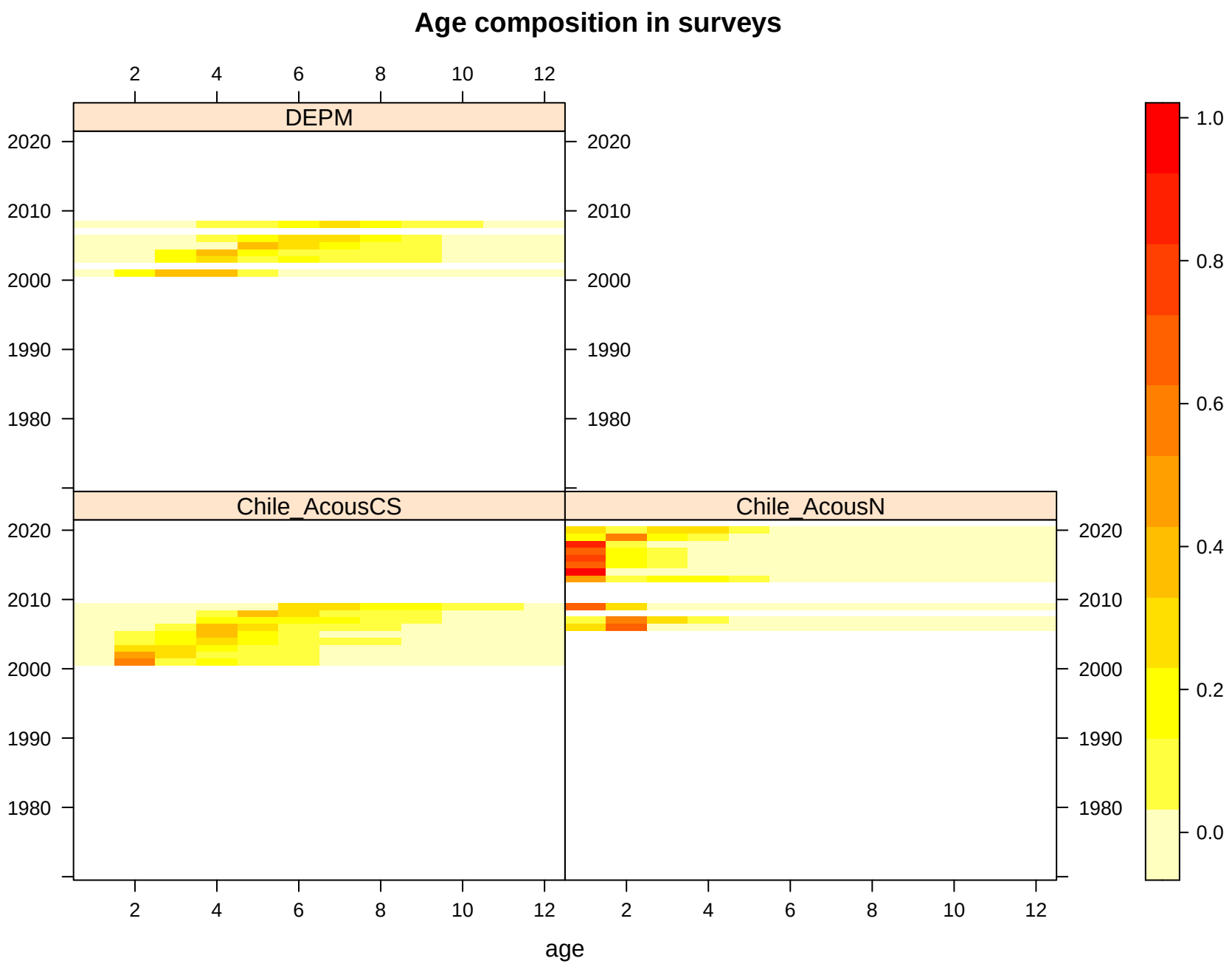
Weight at age by cohort in the fleet

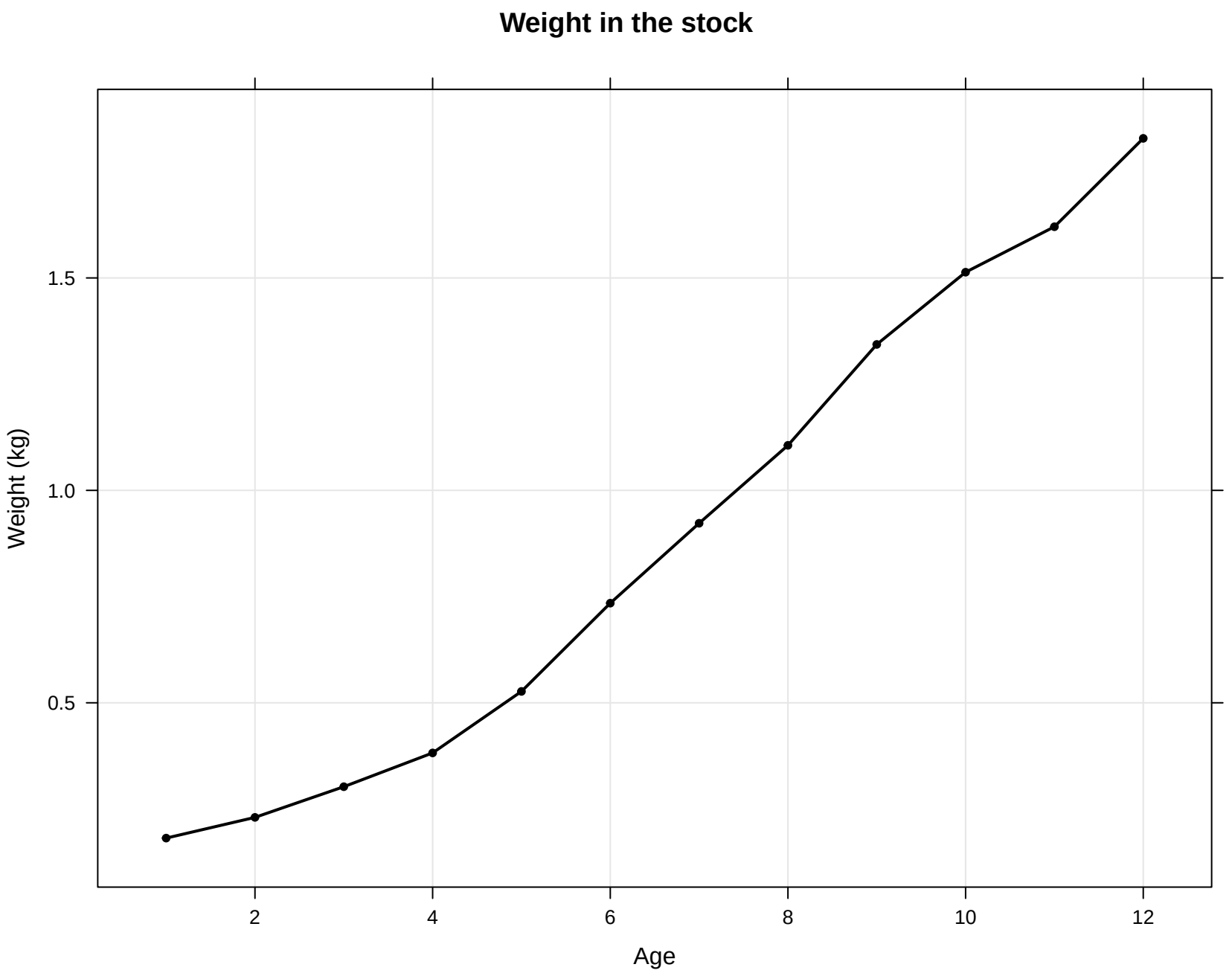


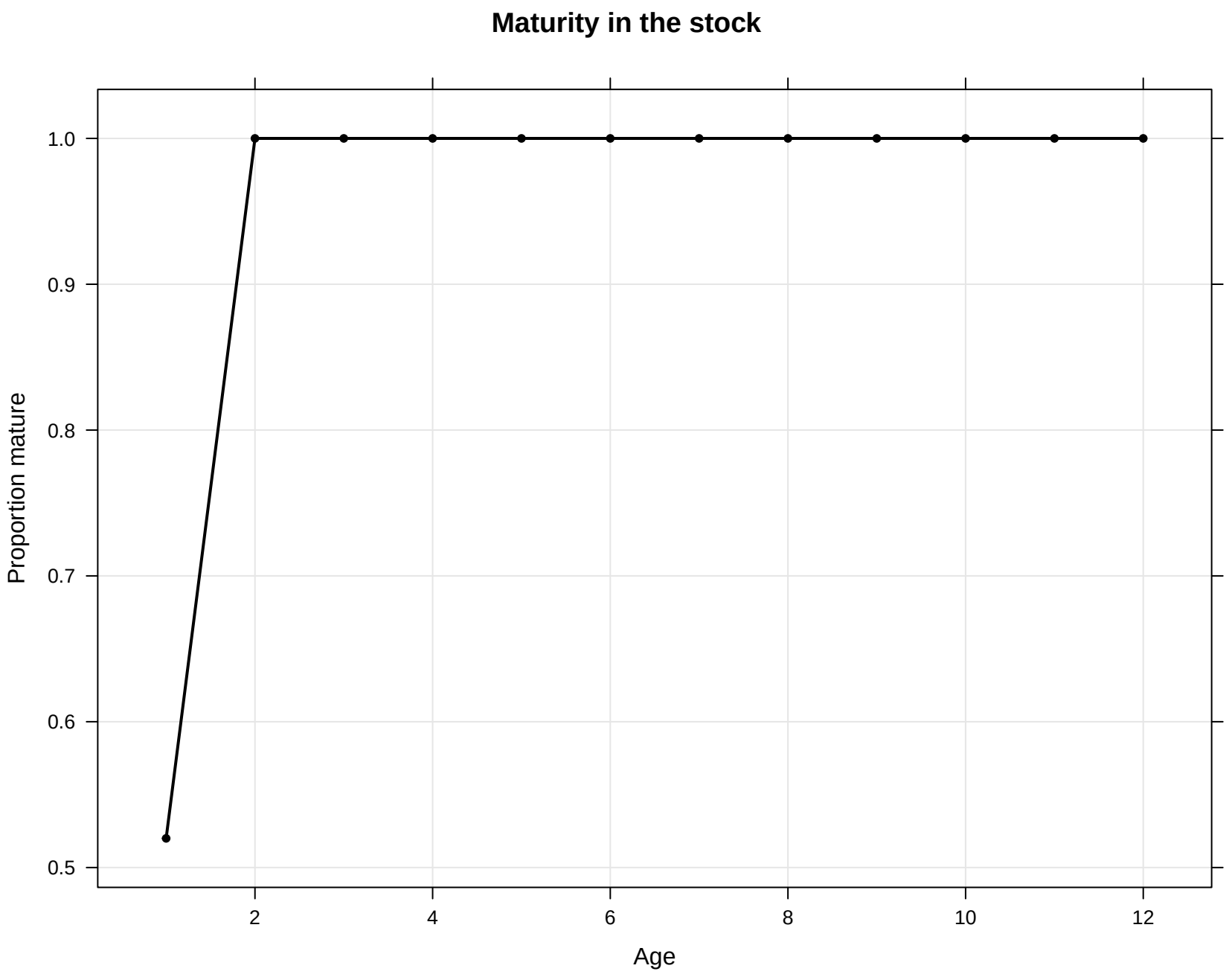

Age composition in fleets Offshore_Trawl
1970
1971
1972
1973
1974
1975
1976
1977
1978
1979
1980
1981
1982
1983
1984
1985
1986
1987
1988
1989
1990
1991
1992
1993
1994
1995
1996
1997
1998
1999
2000
2001
2002
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
Proportion at age
Age
0.0
0.2
0.4
0.6
2 4 6 8 10

Age composition in surveys

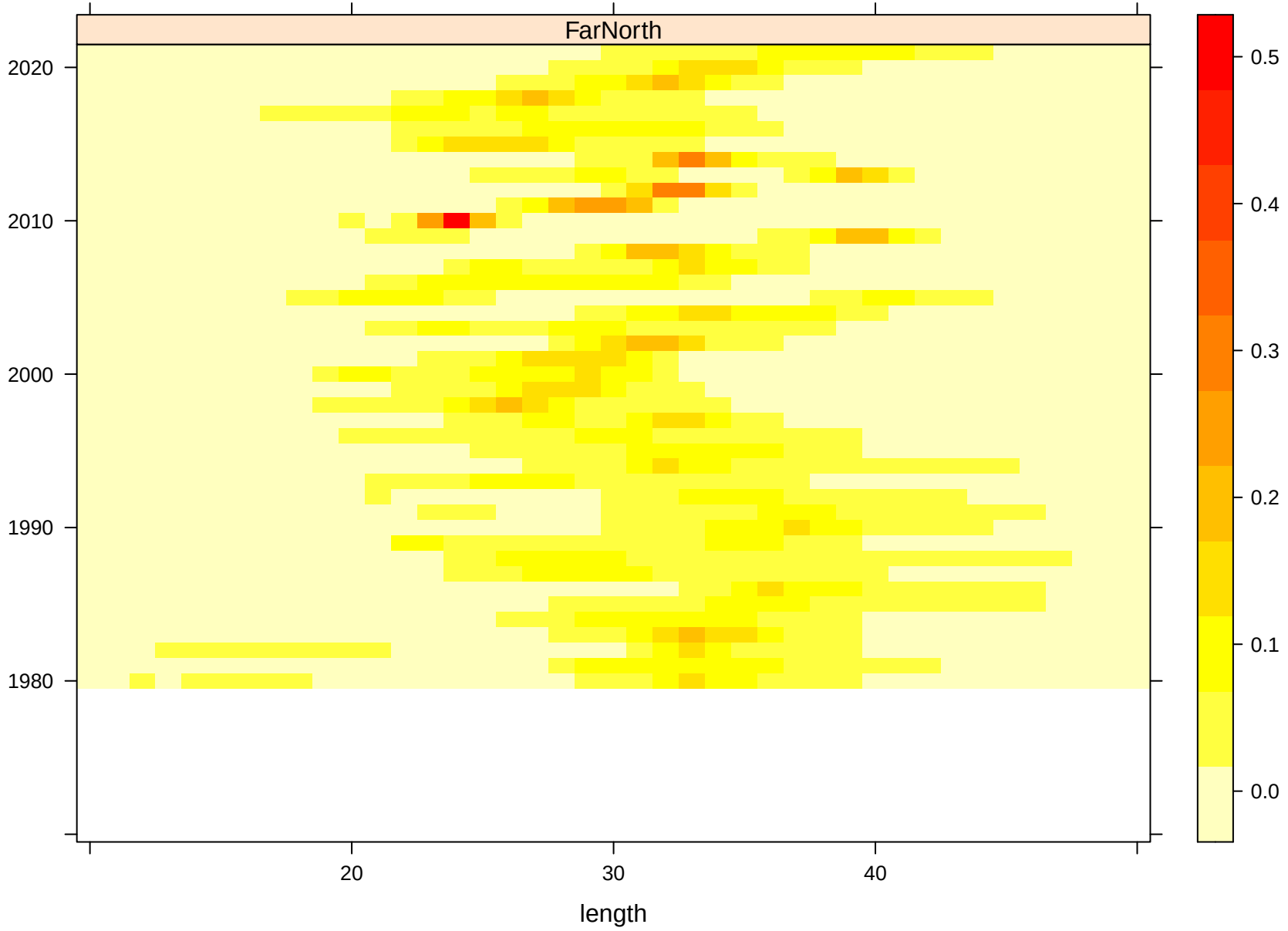


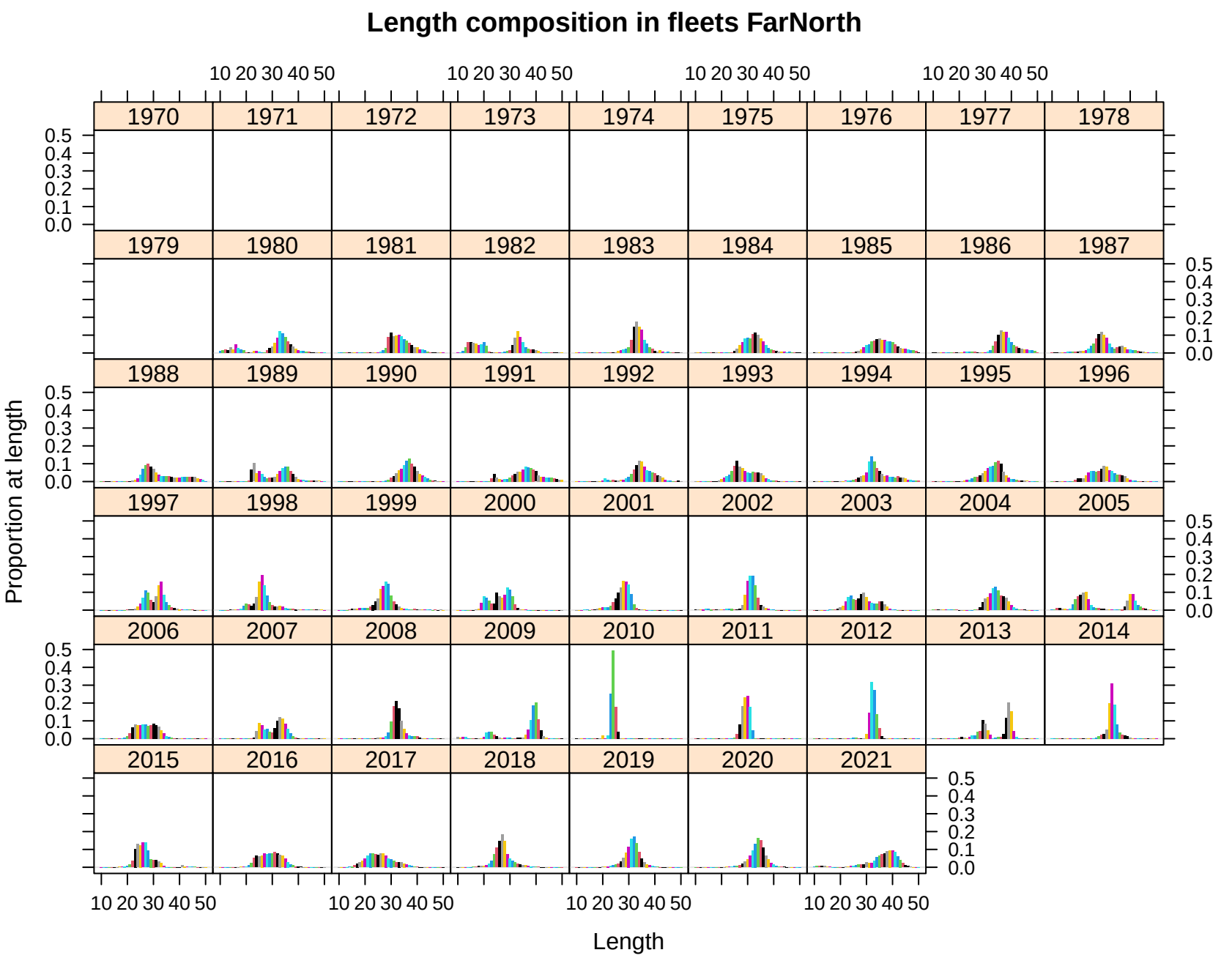

Age composition in surveys
DEPM
Chile_AcousCS
Chile_AcousN
2020
2010
2000
1990
1980
2
4
6
8
10
12
1.0
0.8
0.6
0.4
0.2
0.0
age

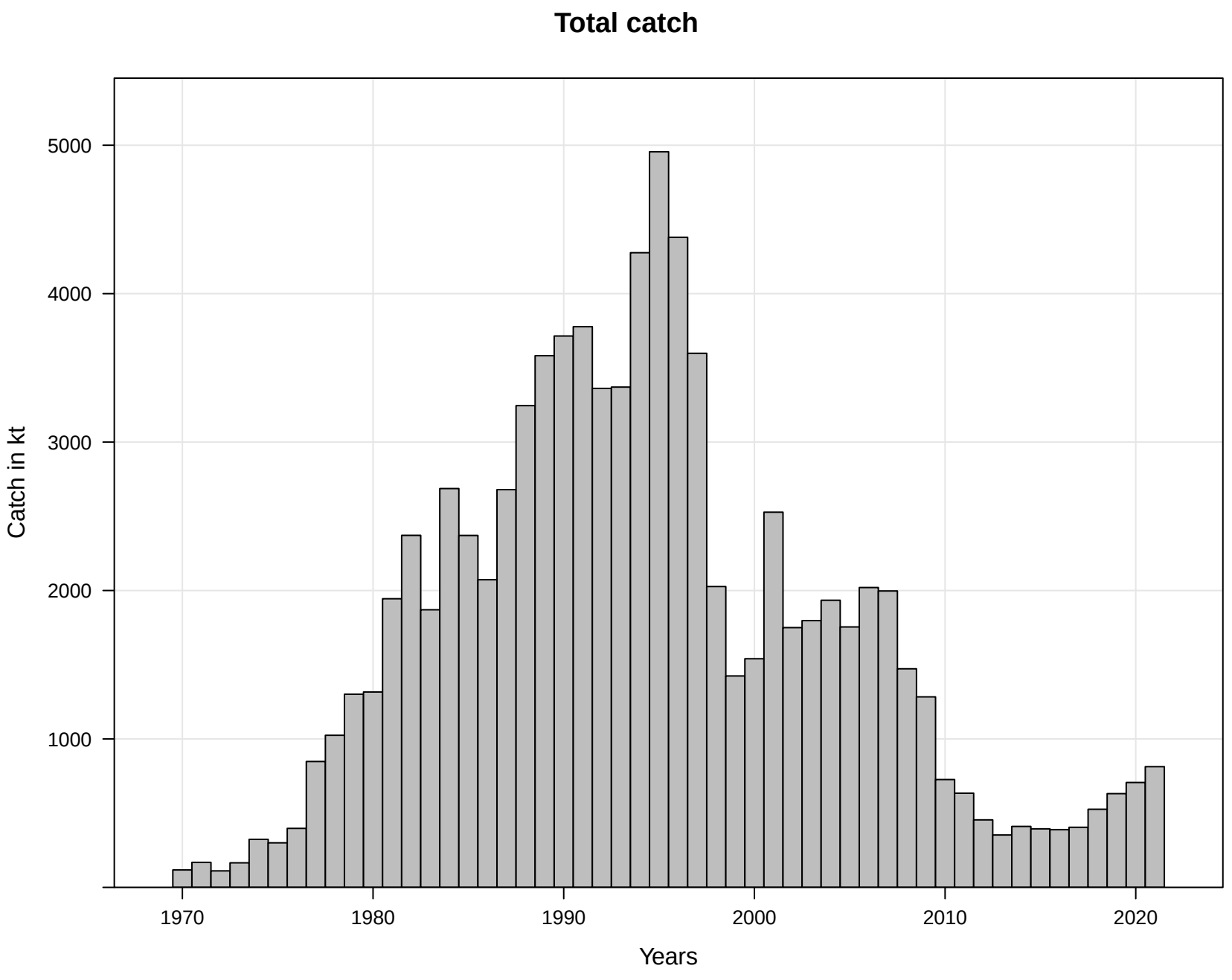

Weight in the stock
Weight (kg)
0.5
1.0
1.5
2
4
6
8
10
12
Age

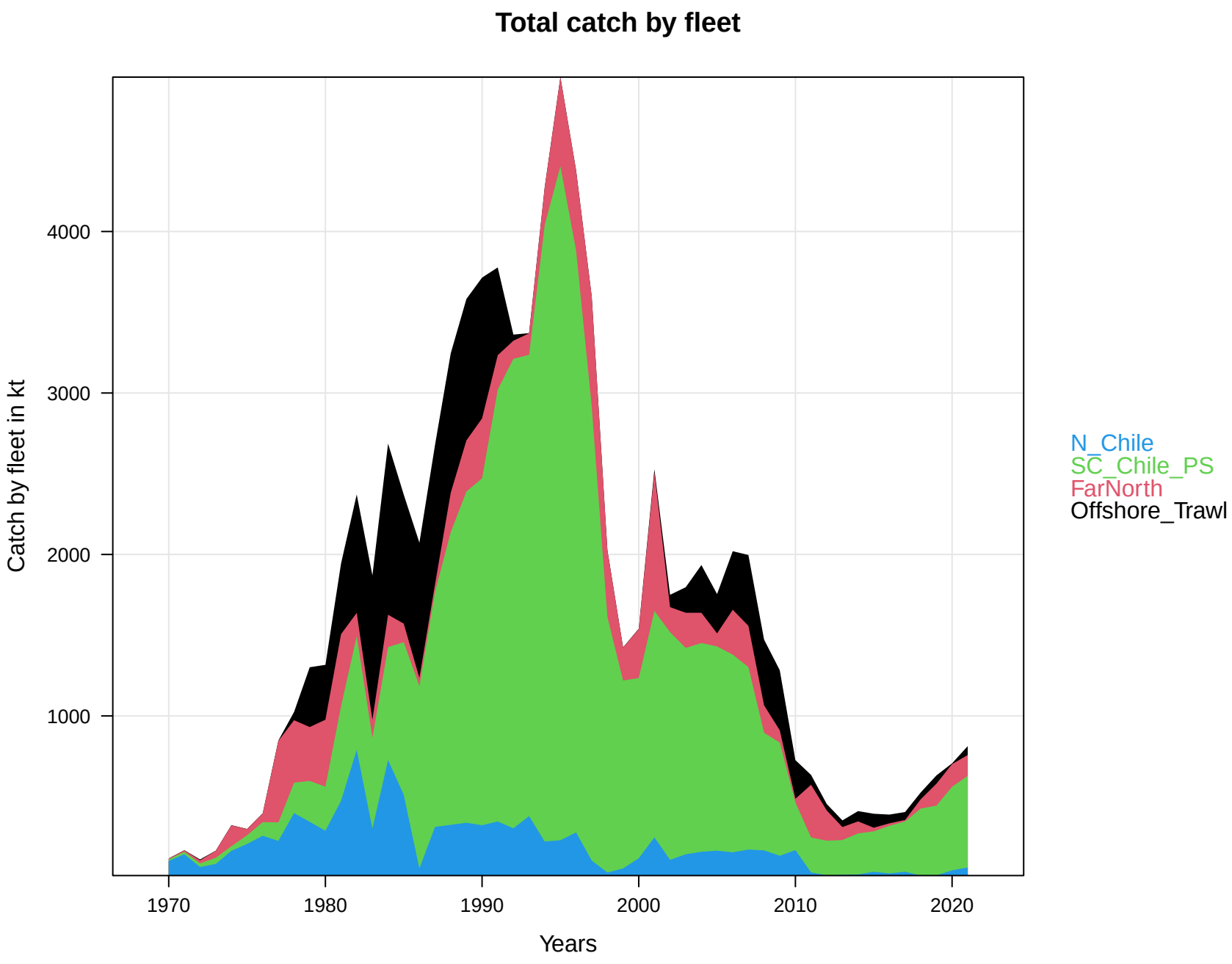

Maturity in the stock
Proportion mature
1.0
0.9
0.8
0.7
0.6
0.5
2
4
6
8
10
12
Age

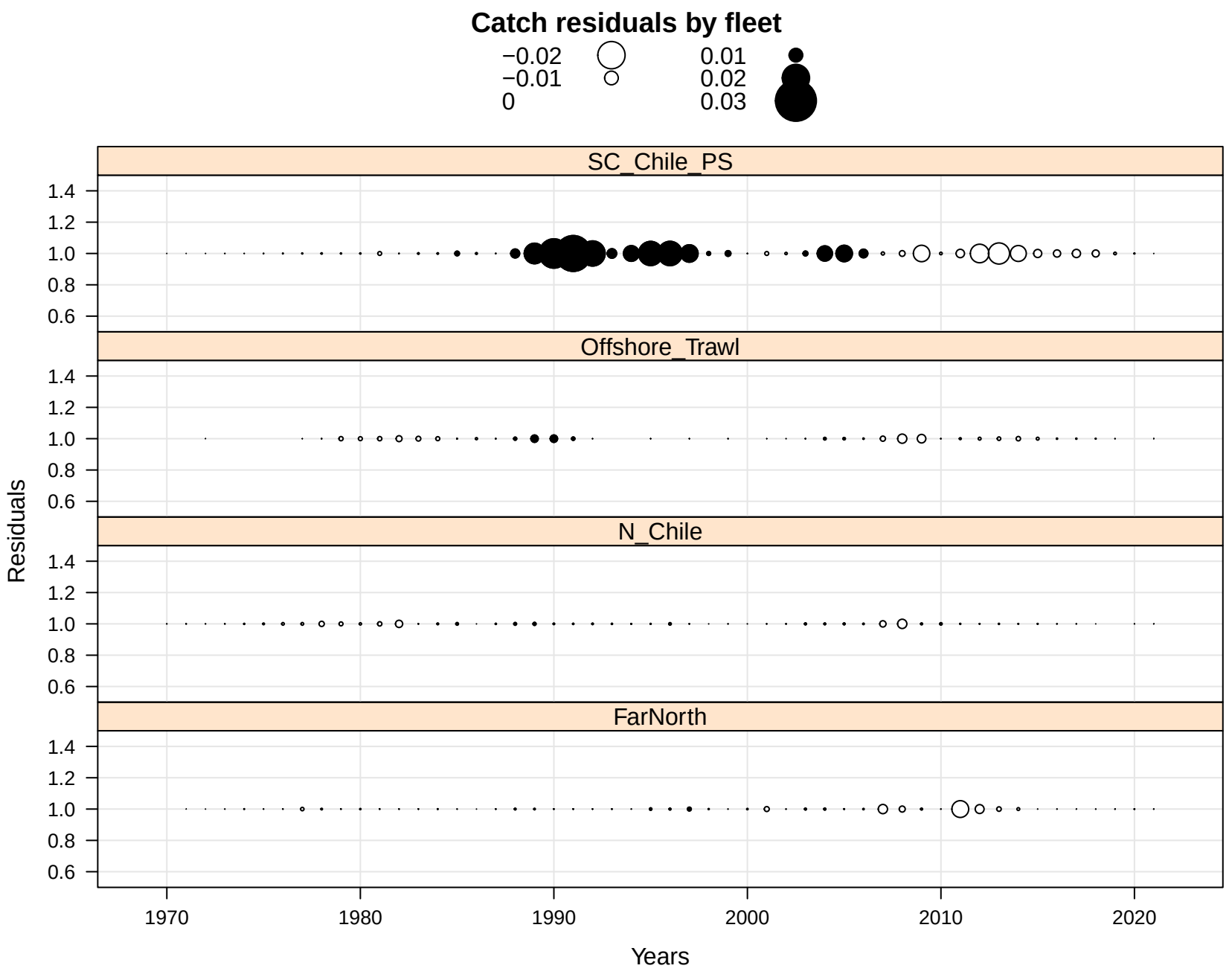
Length composition in fleet FarNorth

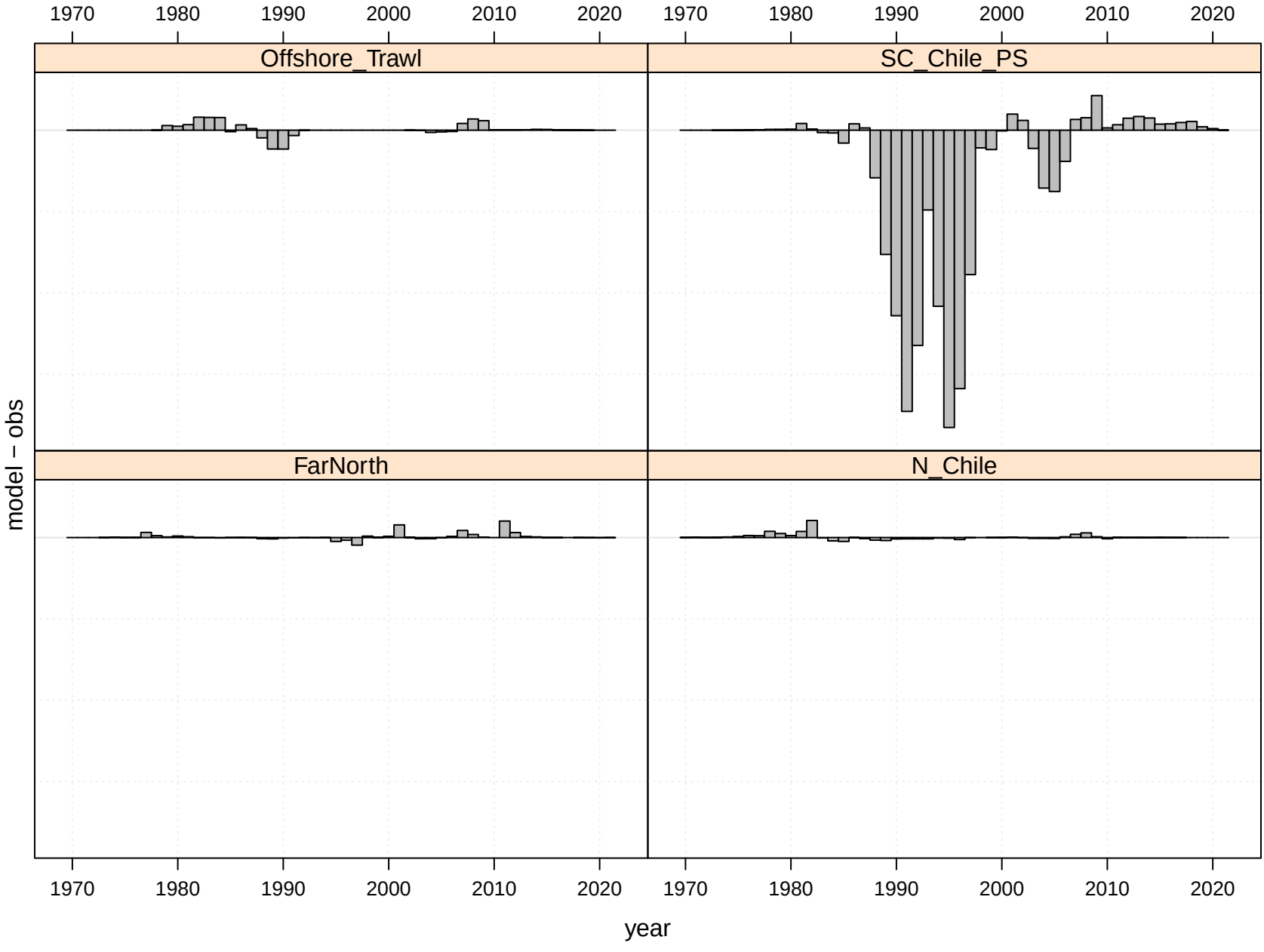


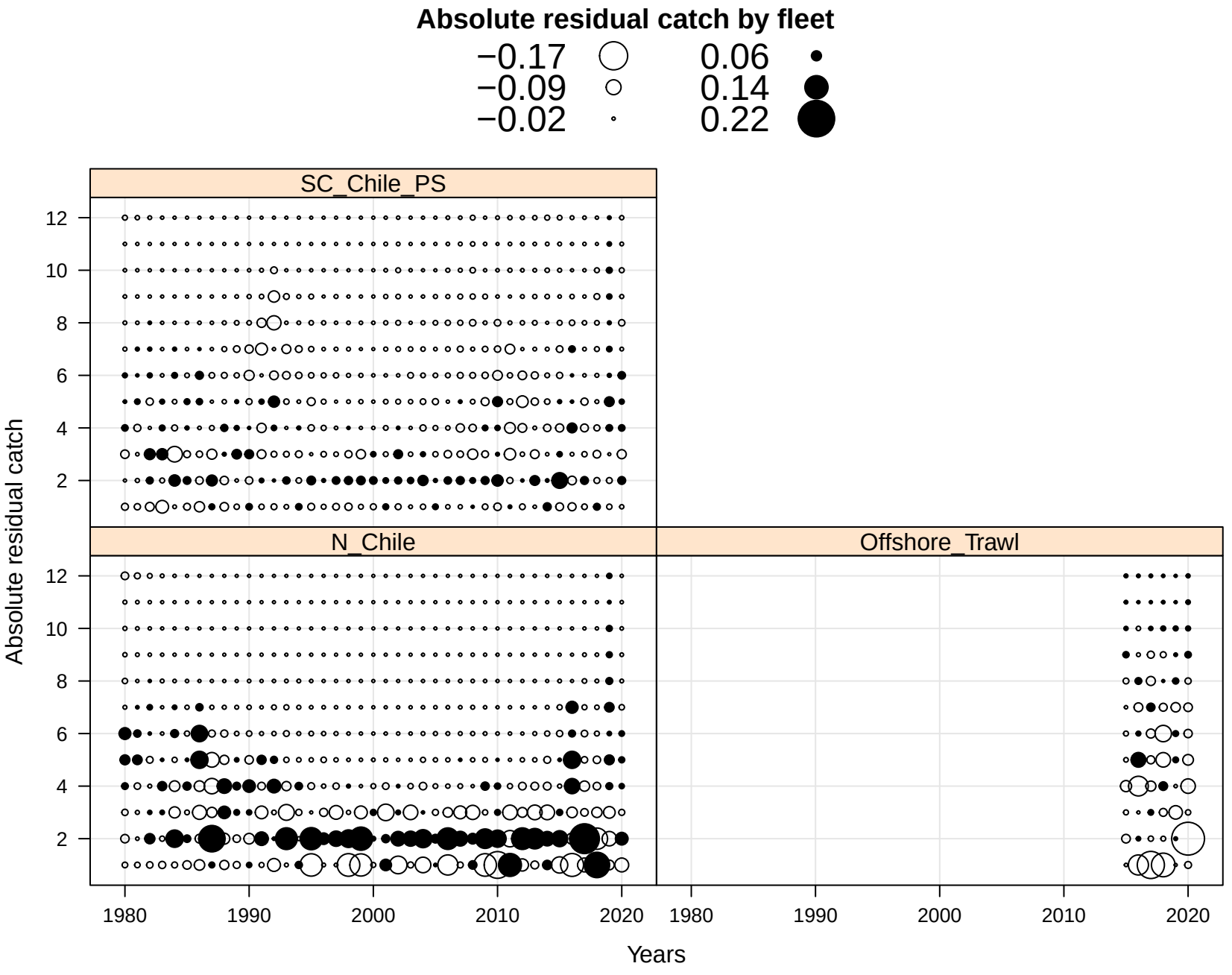

Length composition in fleets FarNorth
1970
1971
1972
1973
1974
1975
1976
1977
1978
1979
1980
1981
1982
1983
1984
1985
1986
1987
1988
1989
1990
1991
1992
1993
1994
1995
1996
1997
1998
1999
2000
2001
2002
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
10 20 30 40 50
0.0
0.1
0.2
0.3
0.4
0.5
Proportion at length
Length

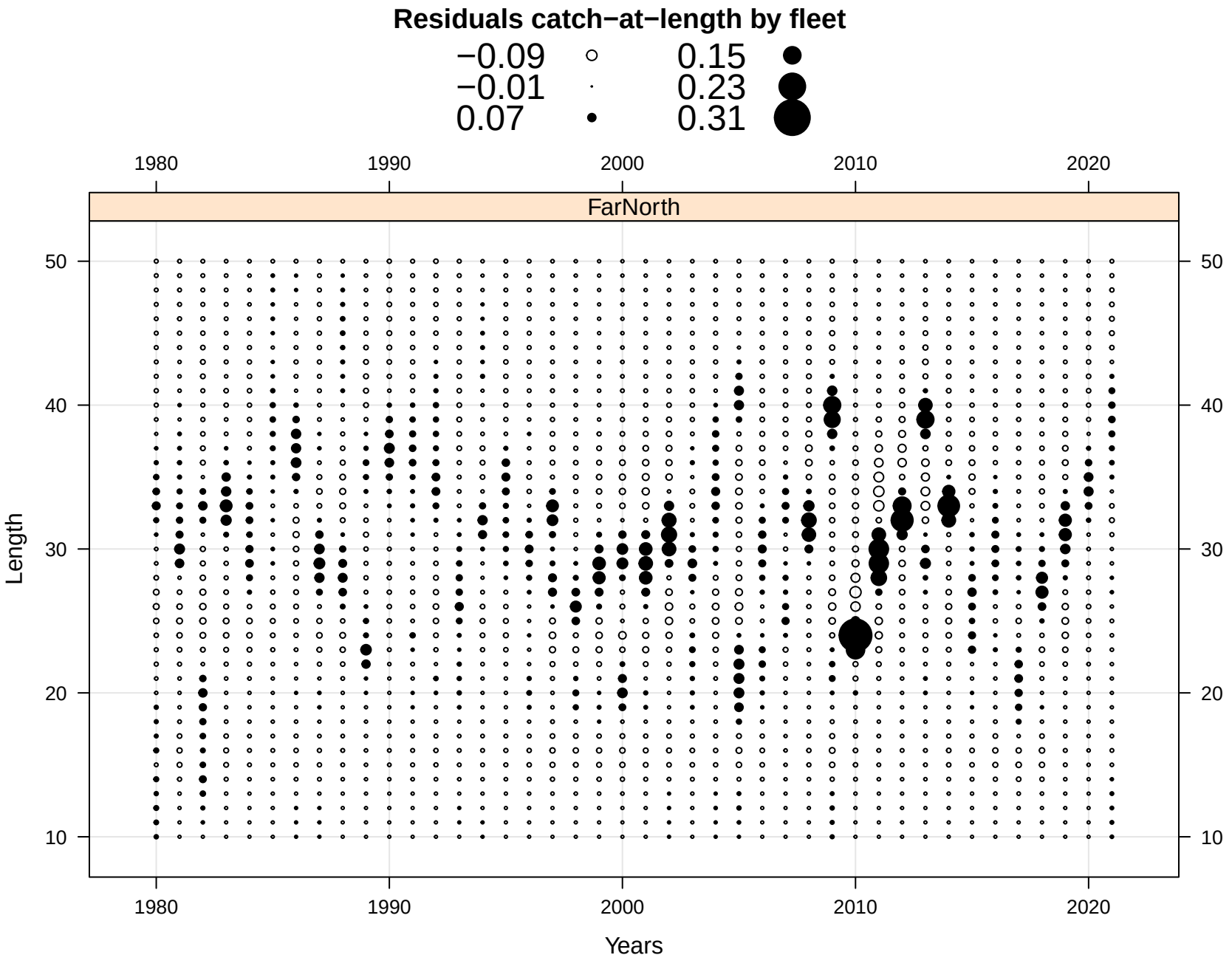

Total catch
Catch in kt
5000
4000
3000
2000
1000
1970
1980
1990
2000
2010
2020
Years

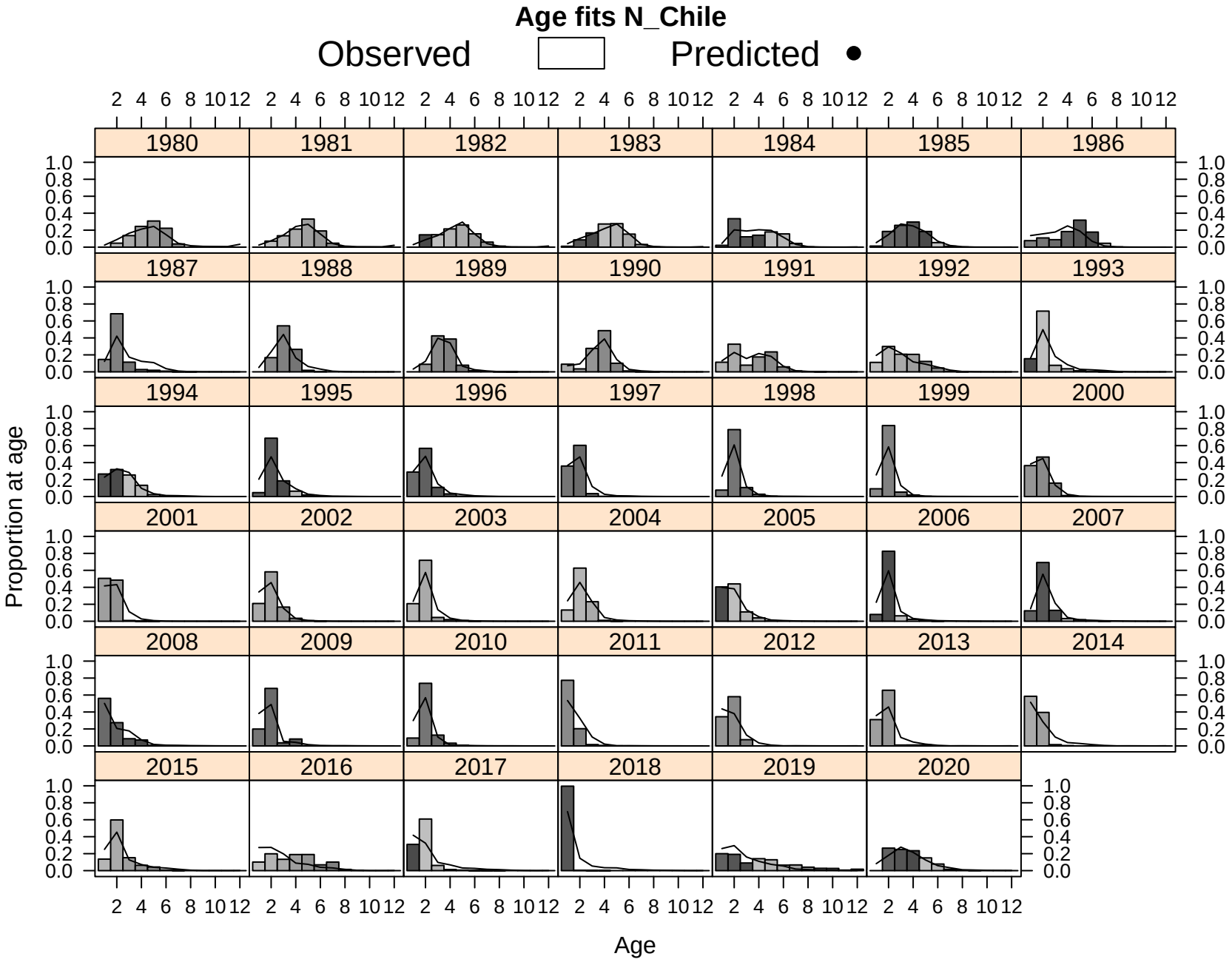

Total catch by fleet
Catch by fleet in kt
4000
3000
2000
1000
1970
1980
1990
2000
2010
2020
Years
N_Chile
SC_Chile_PS
FarNorth
Offshore_Trawl

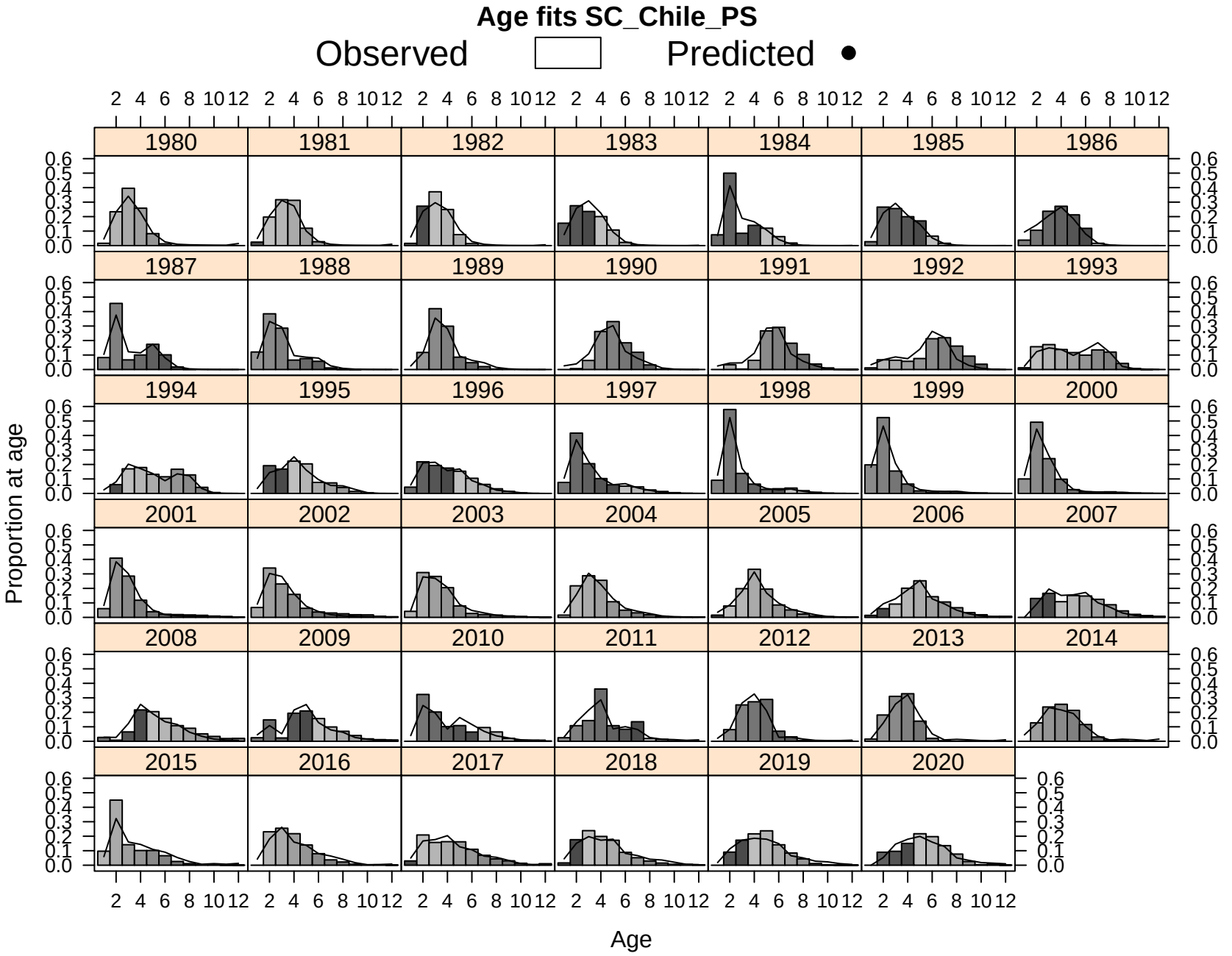

Catch residuals by fleet
−0.02
−0.01
0
0.01
0.02
0.03
SC_Chile_PS
Offshore_Trawl
N_Chile
FarNorth
1.4
1.2
1.0
0.8
0.6
Residuals
1970
1980
1990
2000
2010
2020
Years

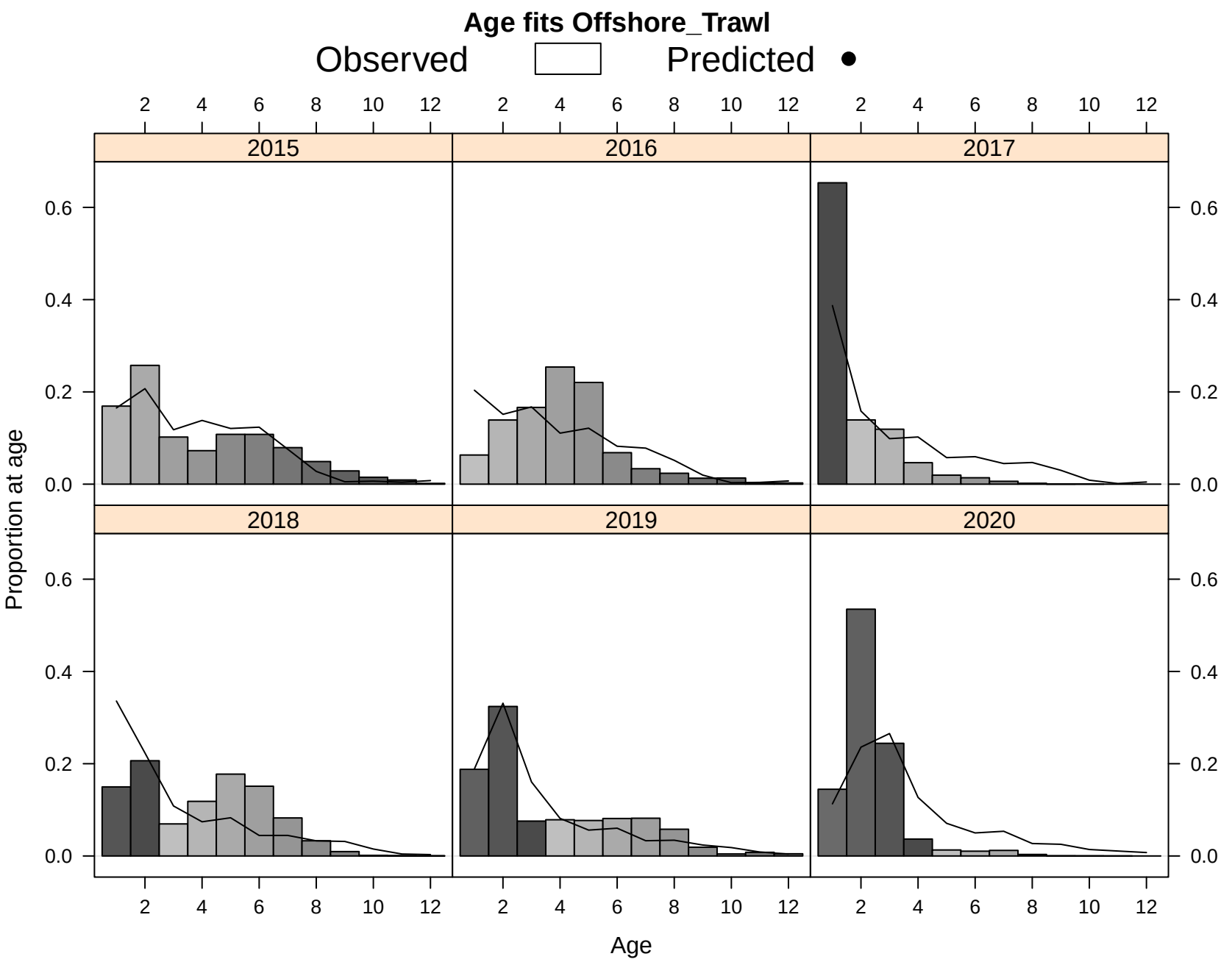

Offshore_Trawl
SC_Chile_PS
FarNorth
N_Chile
model − obs
year
1970
1980
1990
2000
2010
2020

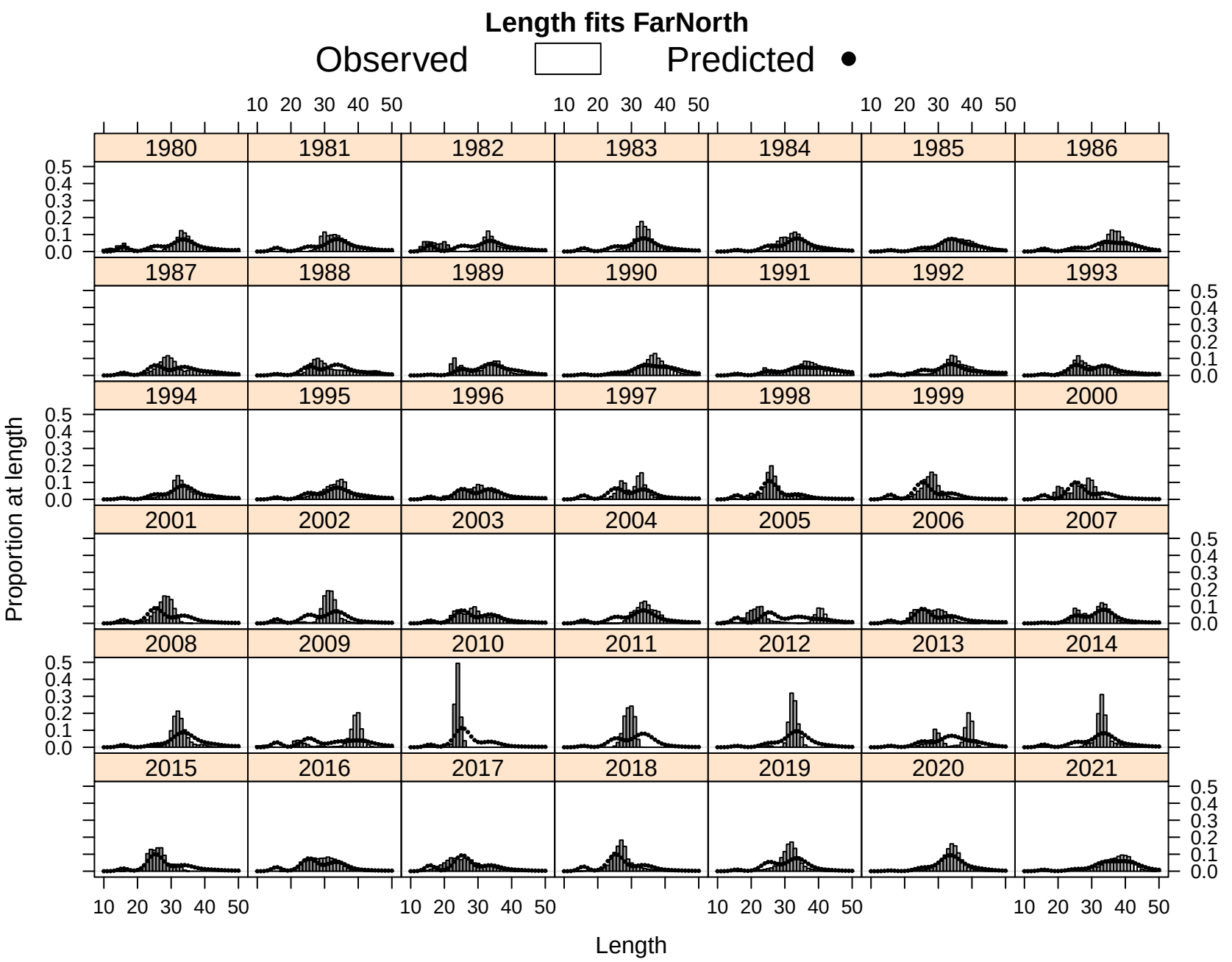

Absolute residual catch by fleet
−0.17
−0.09
−0.02
0.06
0.14
0.22
SC_Chile_PS
N_Chile
Offshore_Trawl
Absolute residual catch
Years
12
10
8
6
4
2
1980
1990
2000
2010
2020

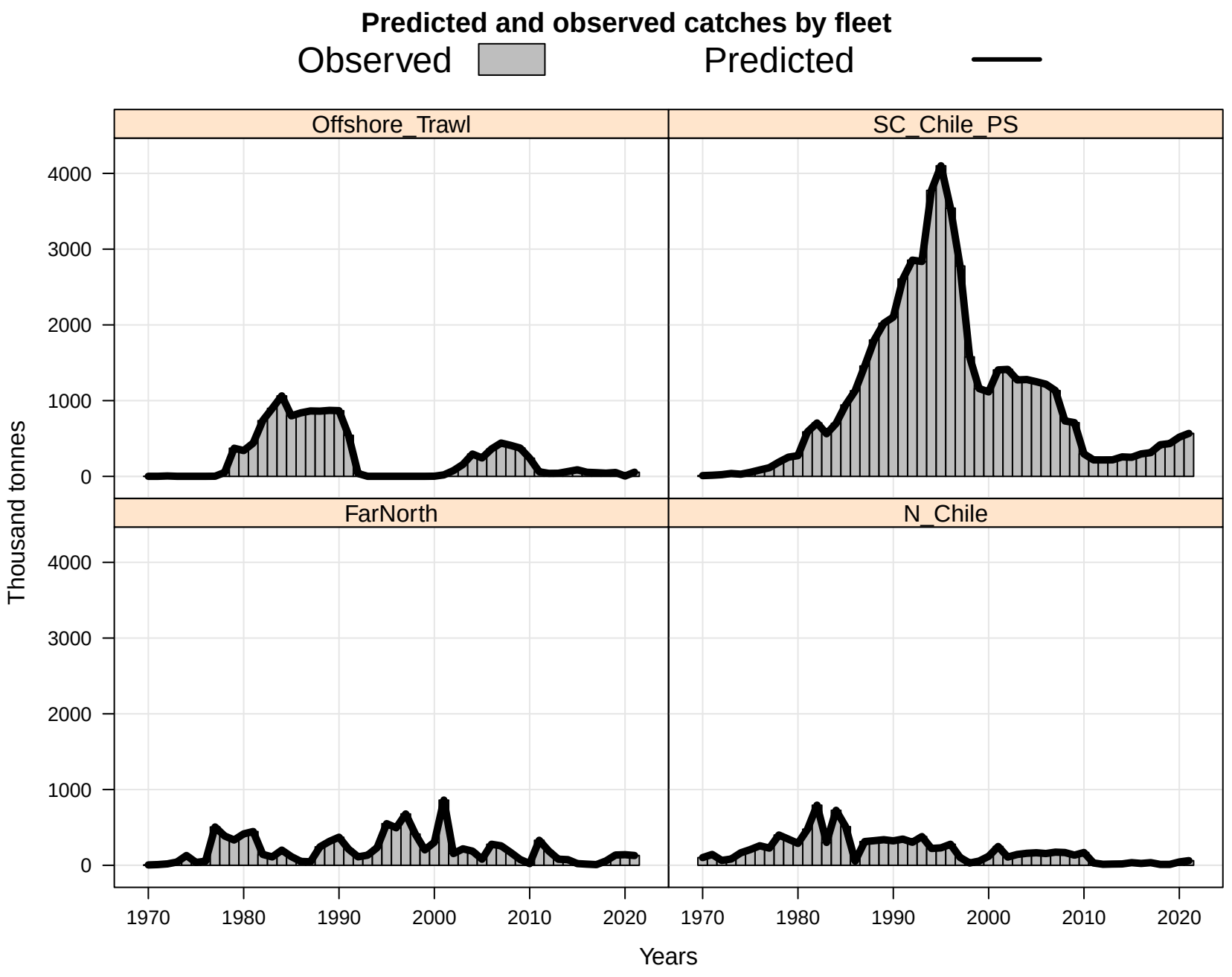

Residuals catch−at−length by fleet
−0.09
−0.01
0.07
0.15
0.23
0.31
FarNorth
Length
Years
1980
1990
2000
2010
2020
10
20
30
40
50

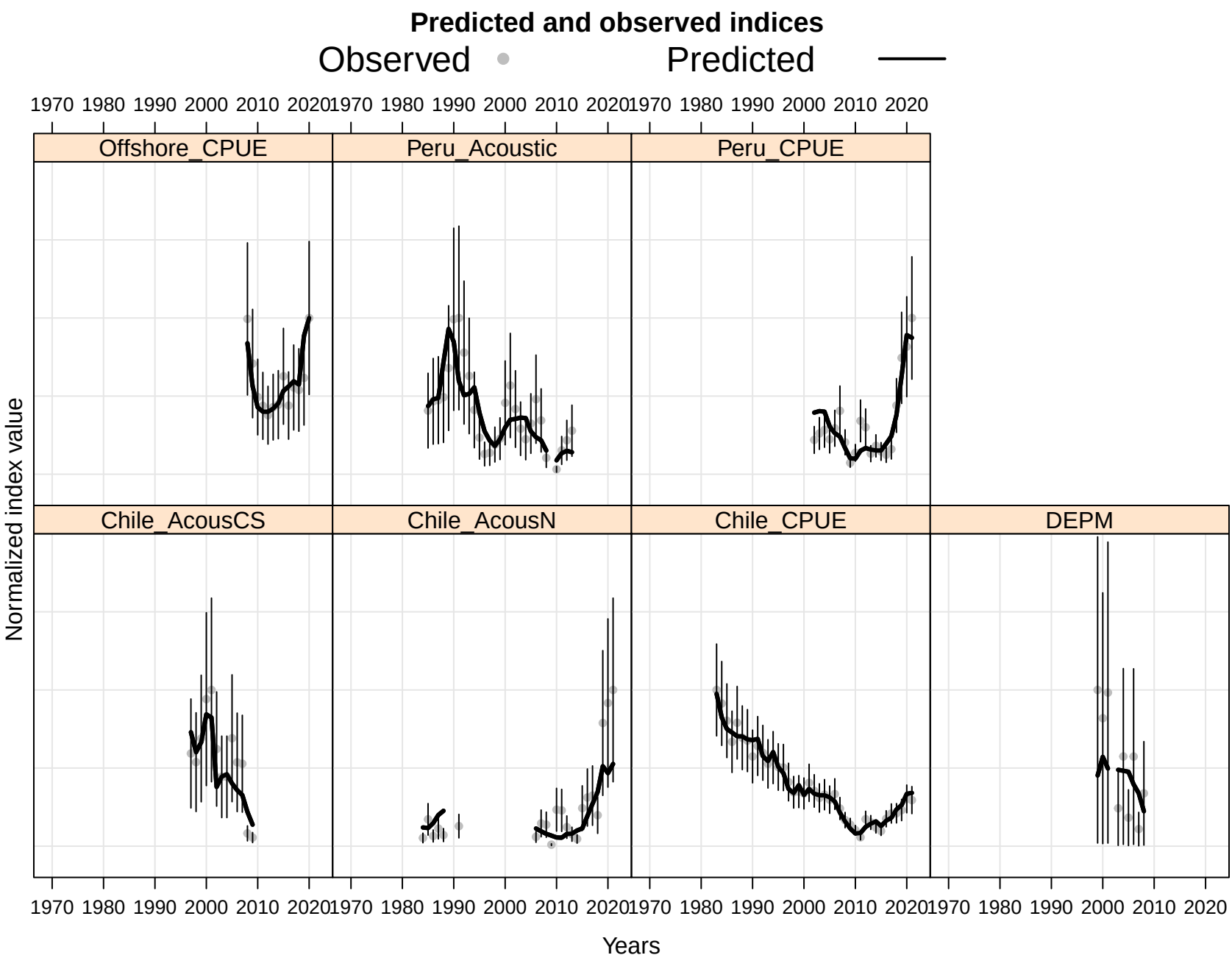

Age fits N_Chile
Observed
Predicted
1980
1981
1982
1983
1984
1985
1986
1987
1988
1989
1990
1991
1992
1993
1994
1995
1996
1997
1998
1999
2000
2001
2002
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
Proportion at age
Age

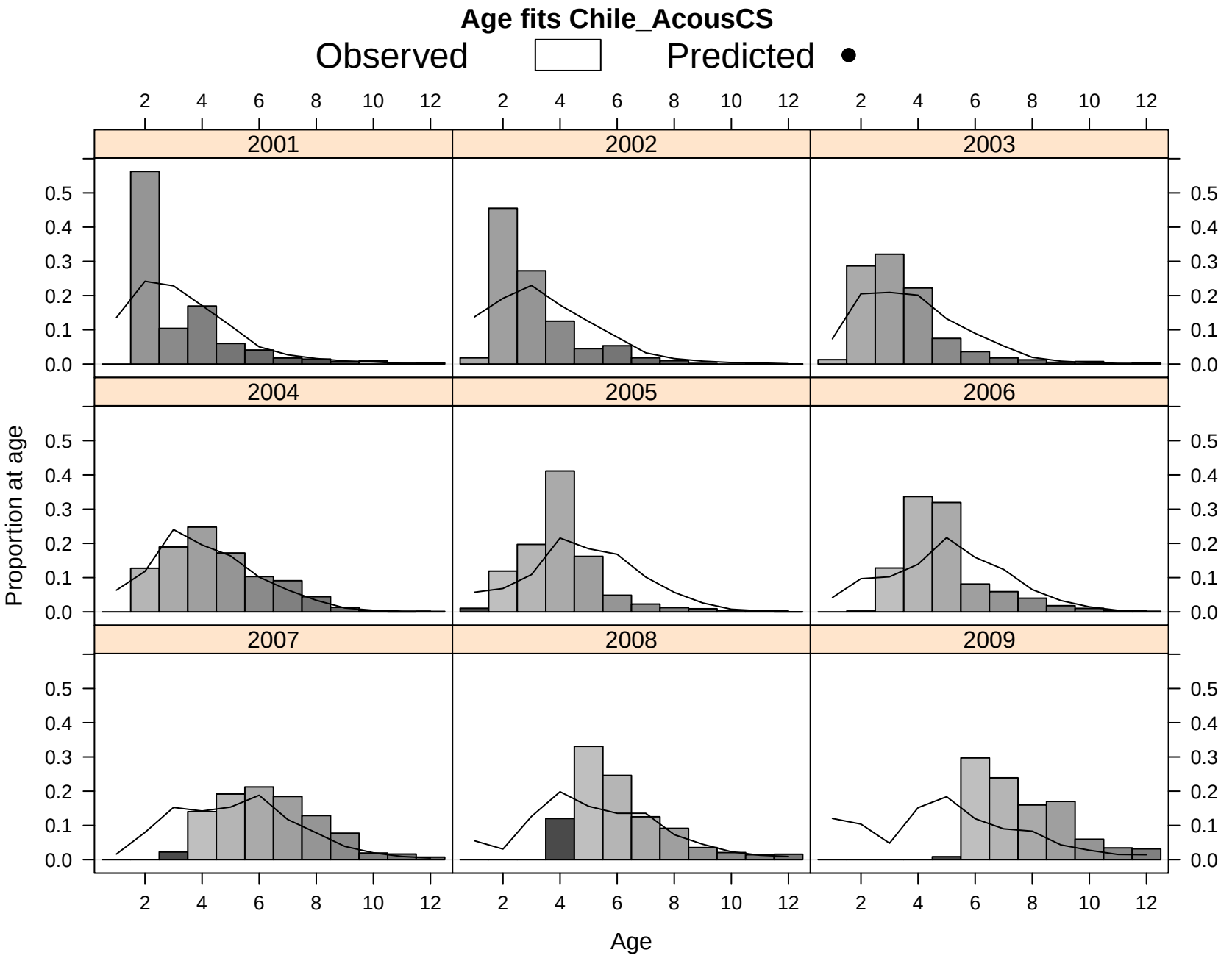

Age fits SC_Chile_PS
Observed
Predicted
1980
1981
1982
1983
1984
1985
1986
1987
1988
1989
1990
1991
1992
1993
1994
1995
1996
1997
1998
1999
2000
2001
2002
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
Proportion at age
Age

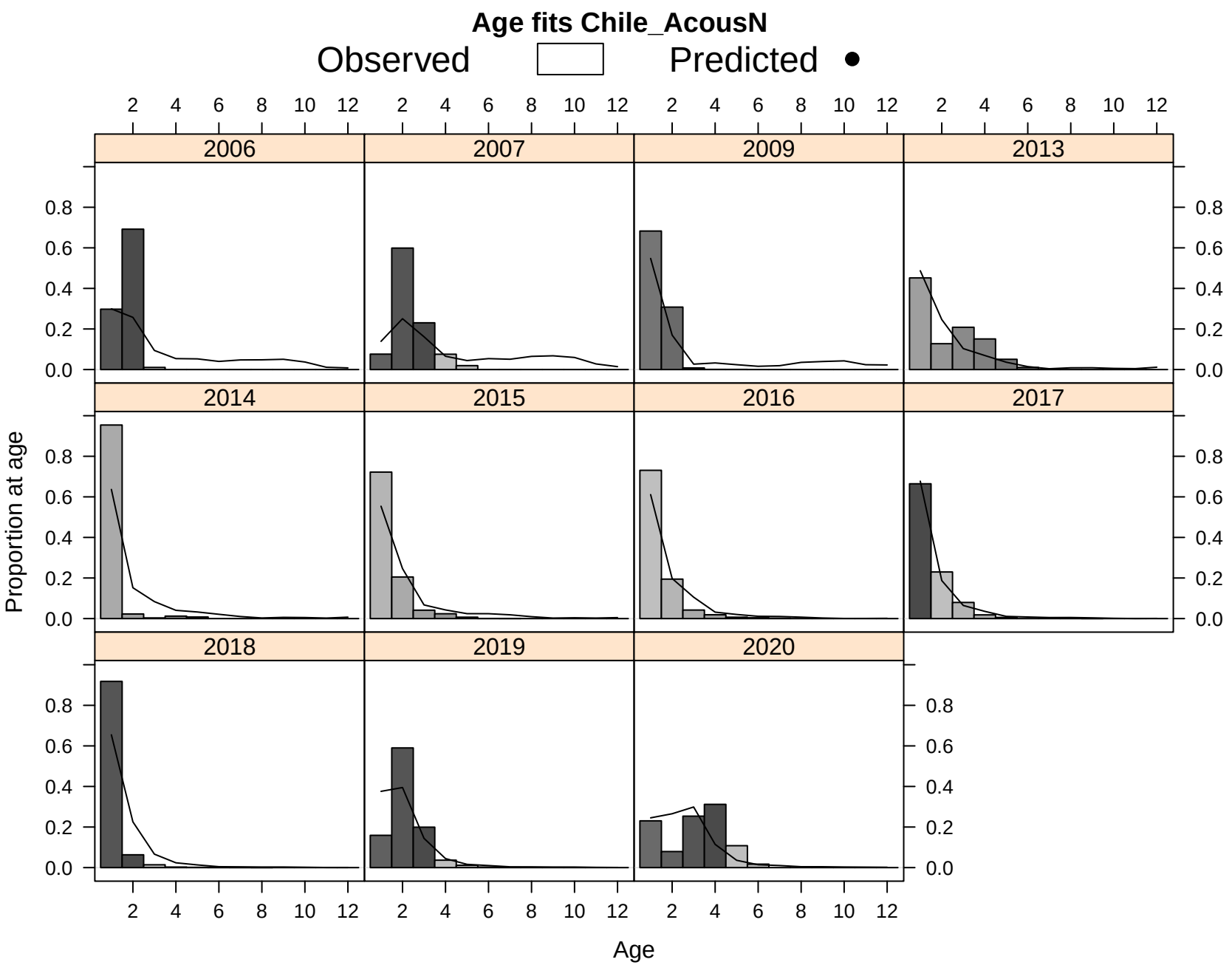

Age fits Offshore_Trawl
Observed
Predicted
2015
2016
2017
2018
2019
2020
Proportion at age
Age
0.0
0.2
0.4
0.6
2
4
6
8
10
12

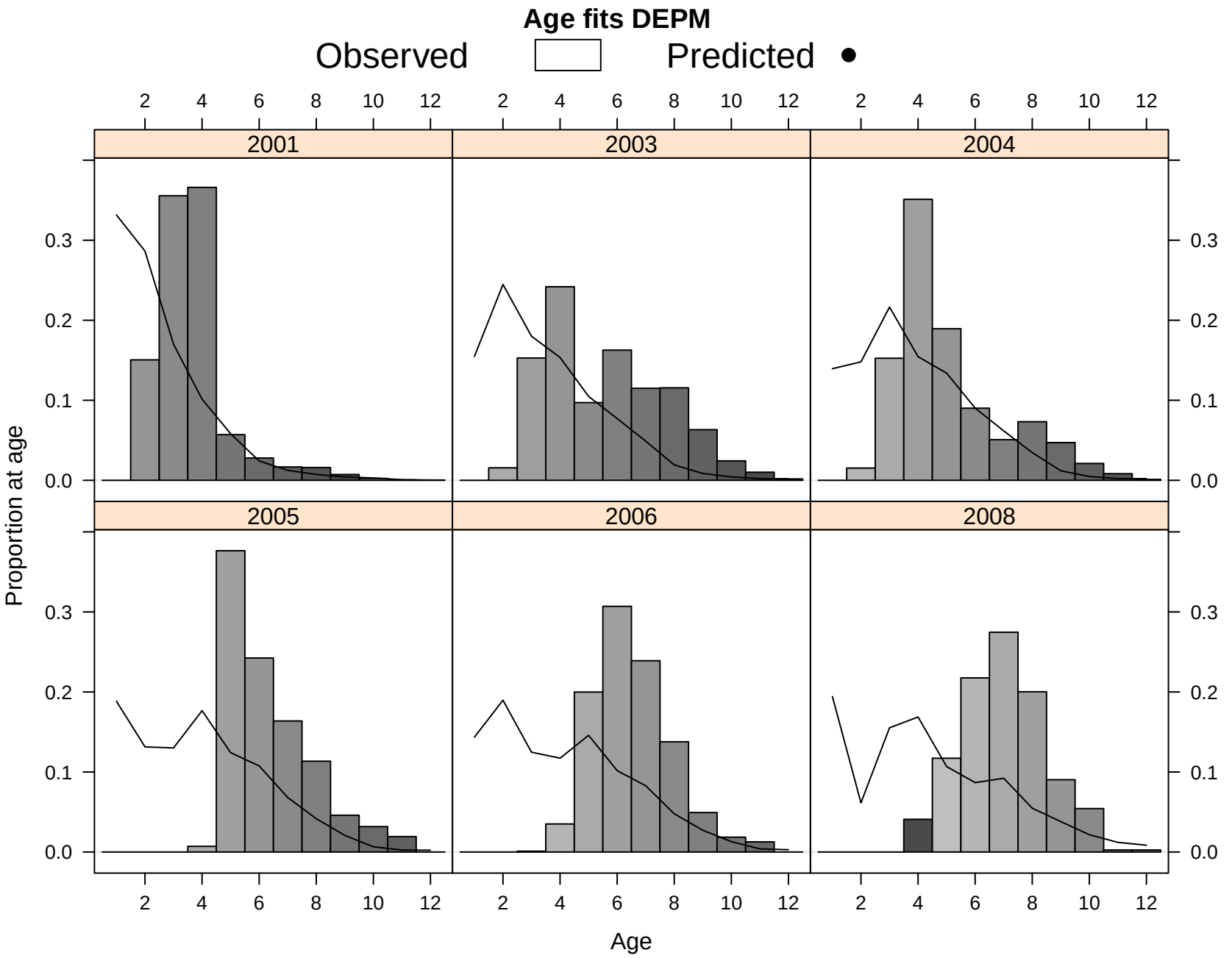

Length fits FarNorth
Observed
Predicted
1980
1981
1982
1983
1984
1985
1986
1987
1988
1989
1990
1991
1992
1993
1994
1995
1996
1997
1998
1999
2000
2001
2002
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
Proportion at length
Length
0.0
0.1
0.2
0.3
0.4
0.5
10
20
30
40
50

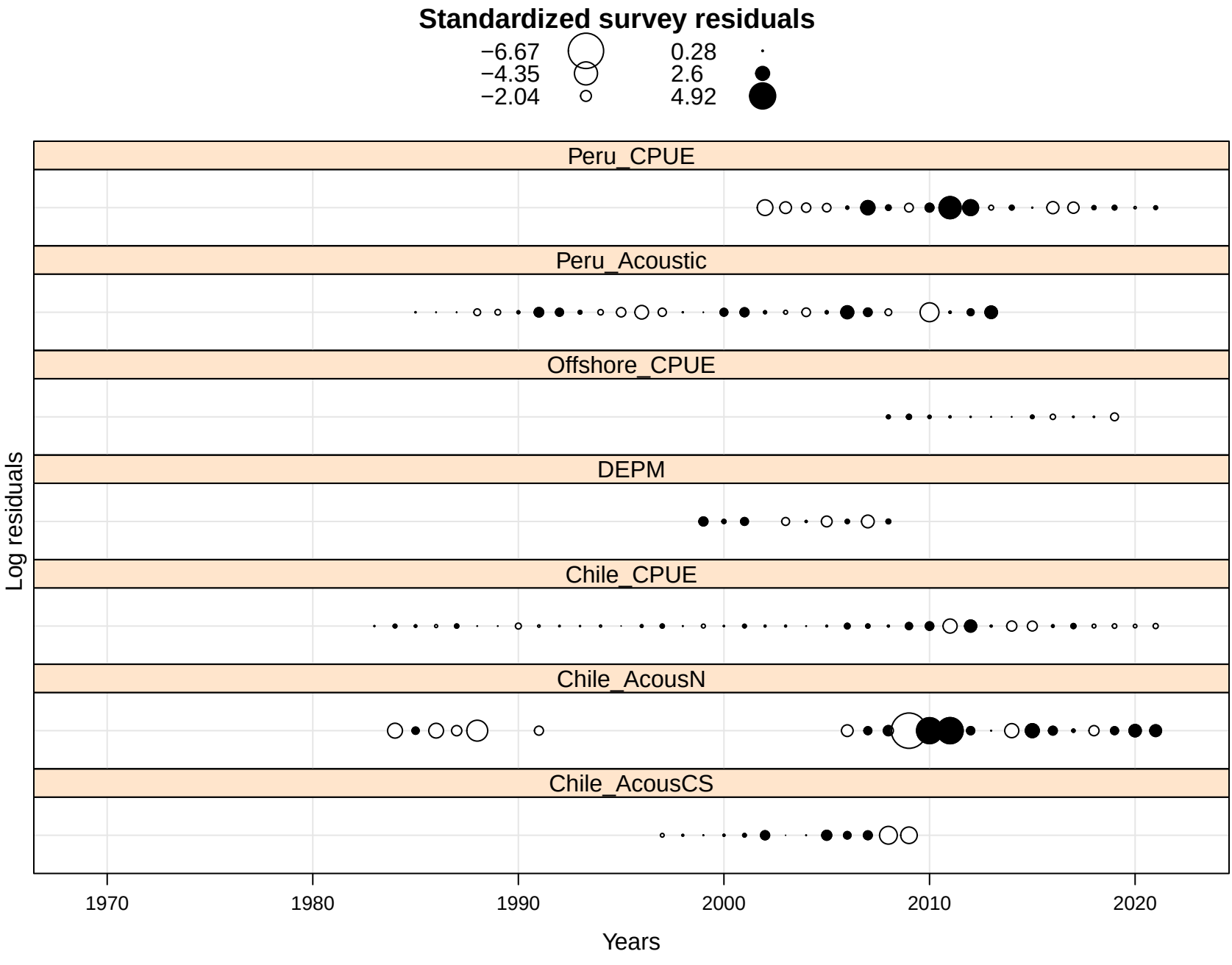

Predicted and observed catches by fleet
Observed
Predicted
Offshore_Trawl
SC_Chile_PS
FarNorth
N_Chile
Thousand tonnes
0
1000
2000
3000
4000
1970
1980
1990
2000
2010
2020
Years

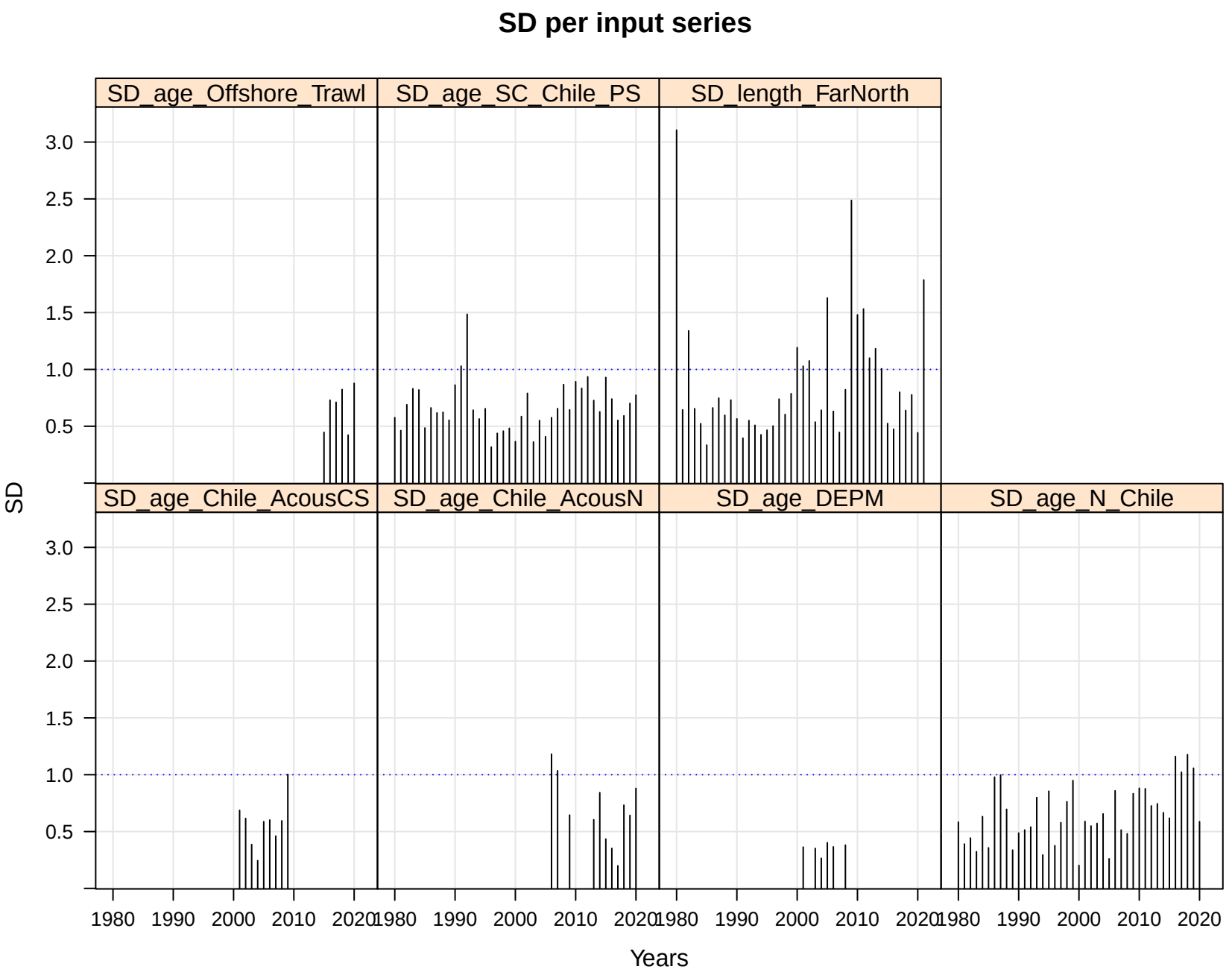

Predicted and observed indices
Observed
Predicted
Offshore_CPUE
Peru_Acoustic
Peru_CPUE
Chile_AcousCS
Chile_AcousN
Chile_CPUE
DEPM
Normalized index value
Years
1970 1980 1990 2000 2010 2020


Age fits Chile_AcousCS
Observed
Predicted
2001
2002
2003
2004
2005
2006
2007
2008
2009
Proportion at age
Age
0.0
0.1
0.2
0.3
0.4
0.5
2
4
6
8
10
12

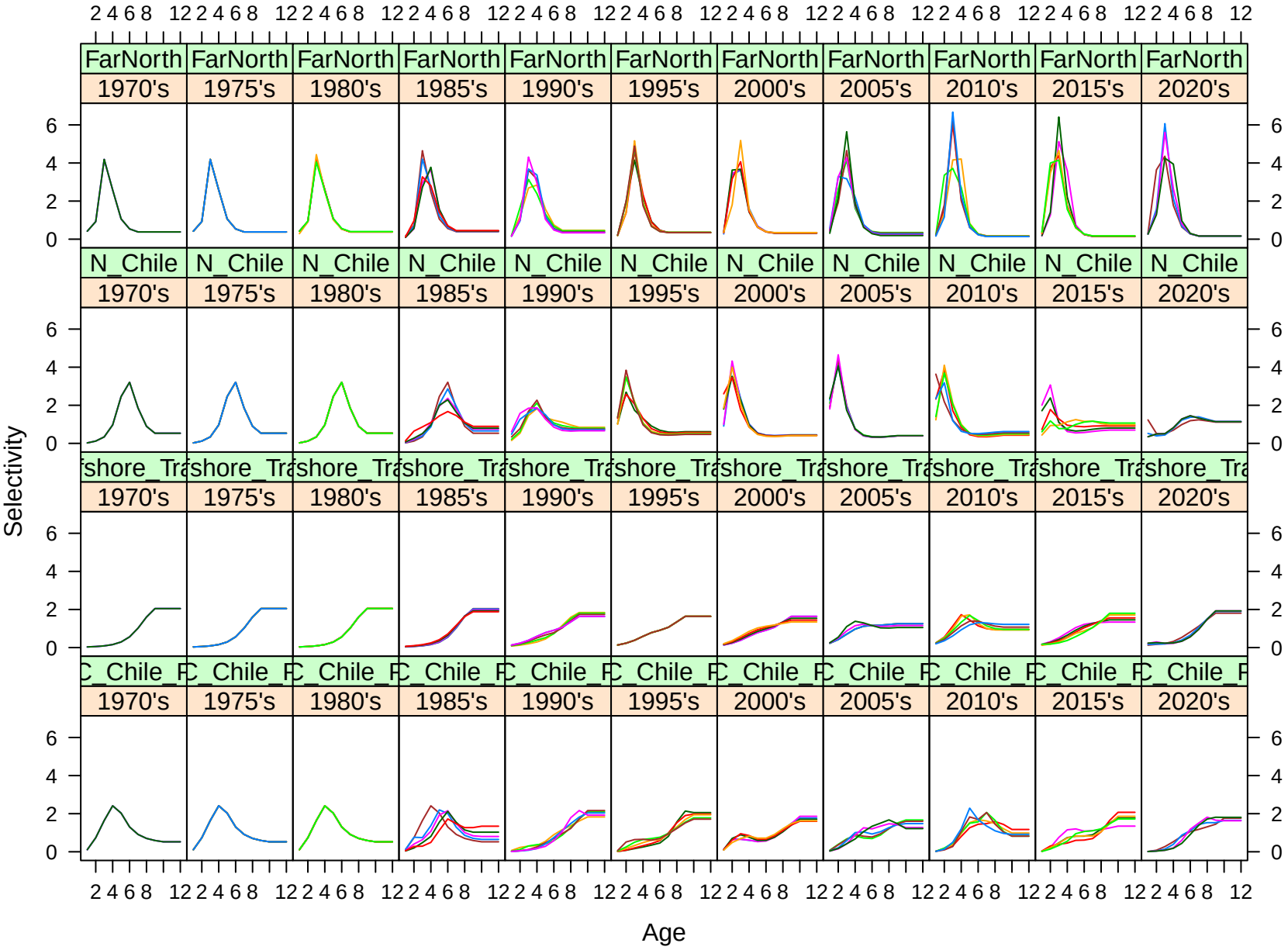

Age fits Chile_AcousN
Observed
Predicted
2006
2007
2009
2013
2014
2015
2016
2017
2018
2019
2020
Proportion at age
Age
0.0
0.2
0.4
0.6
0.8
2
4
6
8
10
12

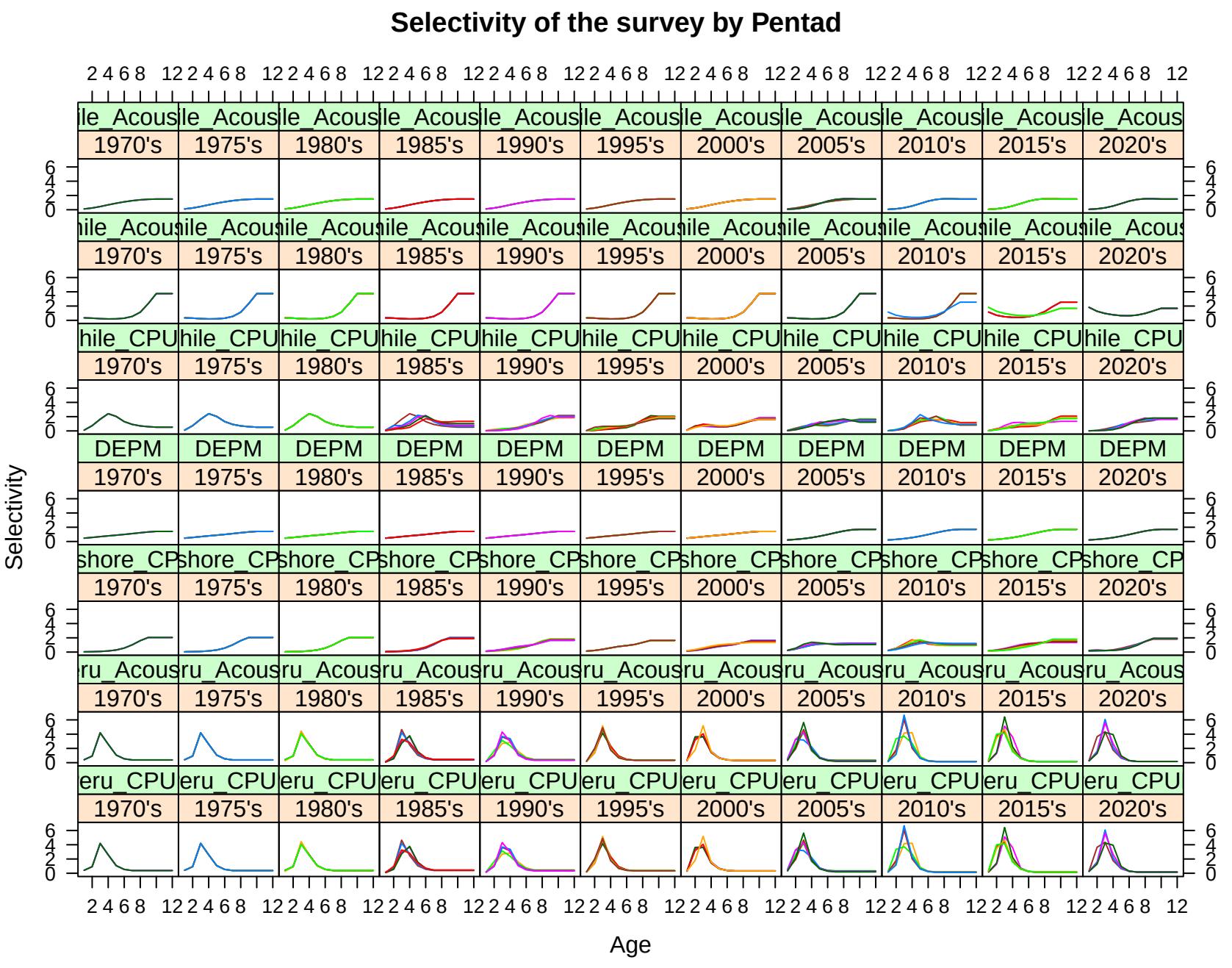

Age fits DEPM
Observed
Predicted
2001
2003
2004
2005
2006
2008
Proportion at age
Age
0.0
0.1
0.2
0.3
2
4
6
8
10
12

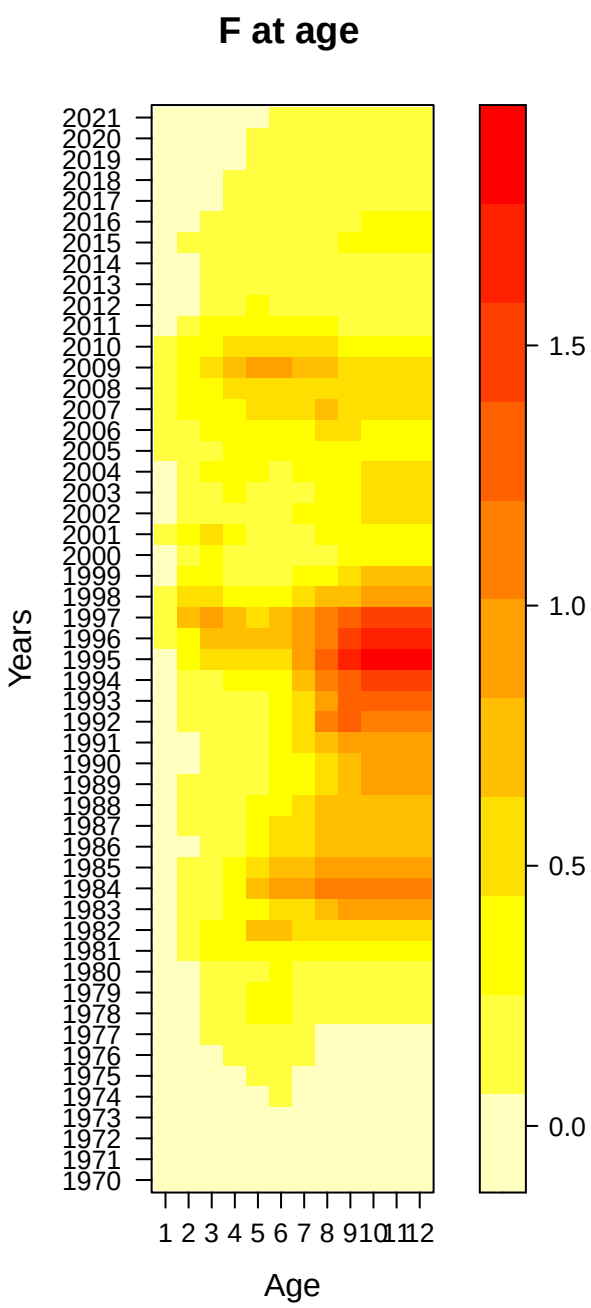

Standardized survey residuals
−6.67
−4.35
−2.04
0.28
2.6
4.92
Peru_CPUE
Peru_Acoustic
Offshore_CPUE
DEPM
Chile_CPUE
Chile_AcousN
Chile_AcousCS
Log residuals
1970
1980
1990
2000
2010
2020
Years


SD per input series
SD_age_Offshore_Trawl
SD_age_SC_Chile_PS
SD_length_FarNorth
SD_age_Chile_AcousCS
SD_age_Chile_AcousN
SD_age_DEPM
SD_age_N_Chile
SD
Years
0.5
1.0
1.5
2.0
2.5
3.0
1980
1990
2000
2010
2020

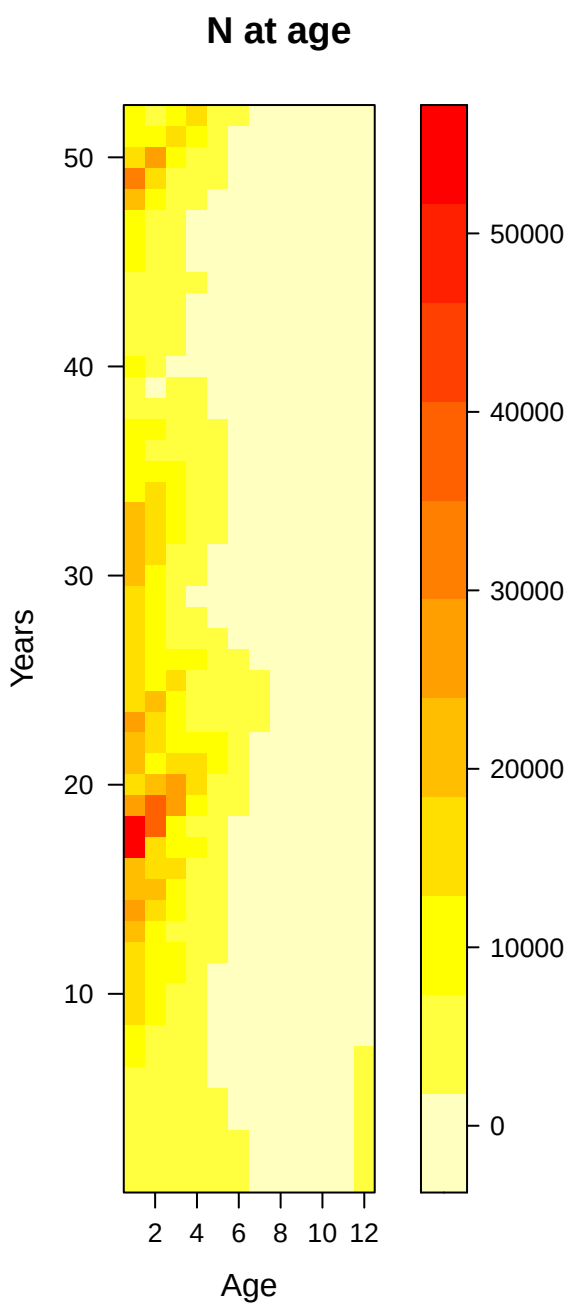
Selectivity of the Fishery by Pentad

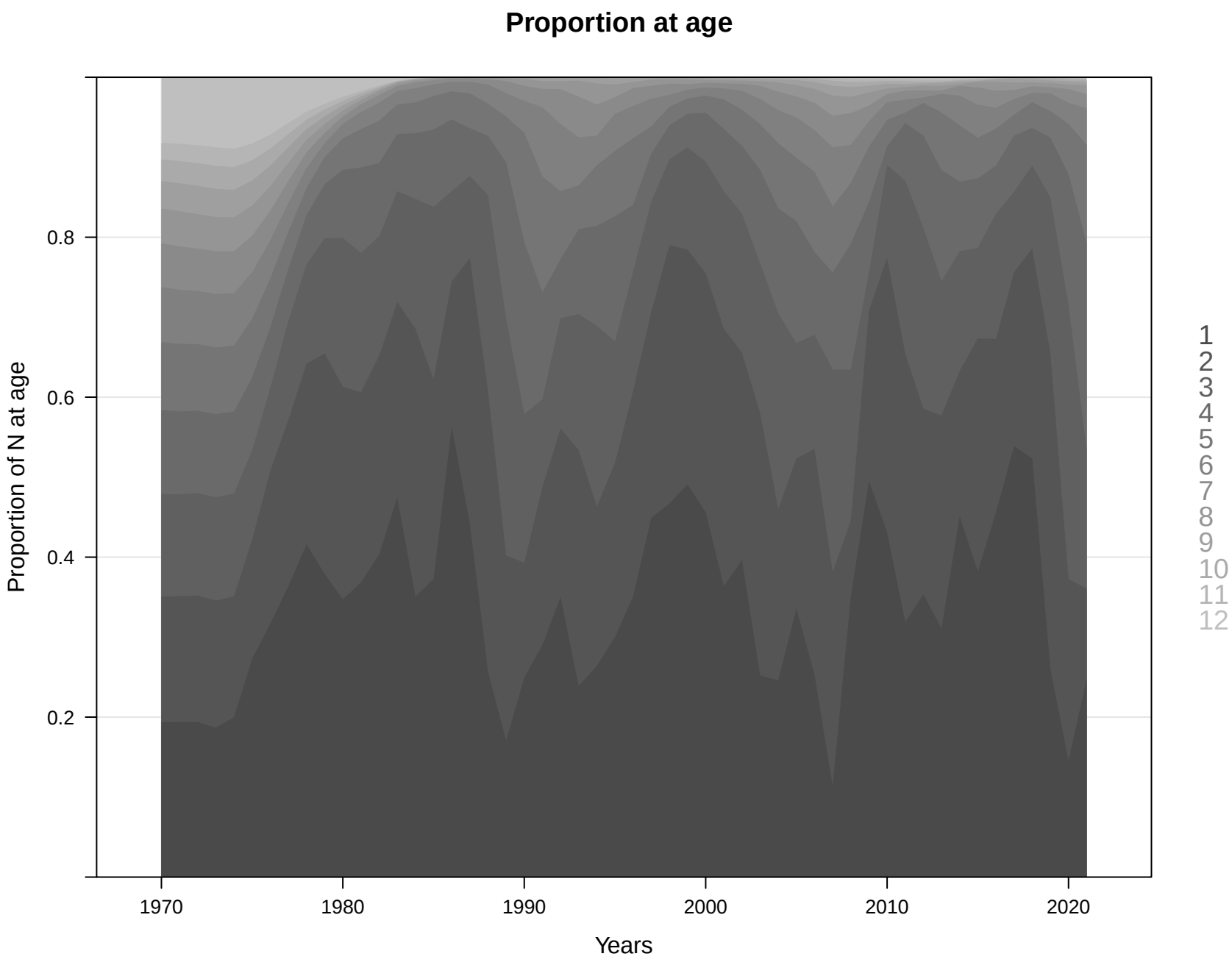


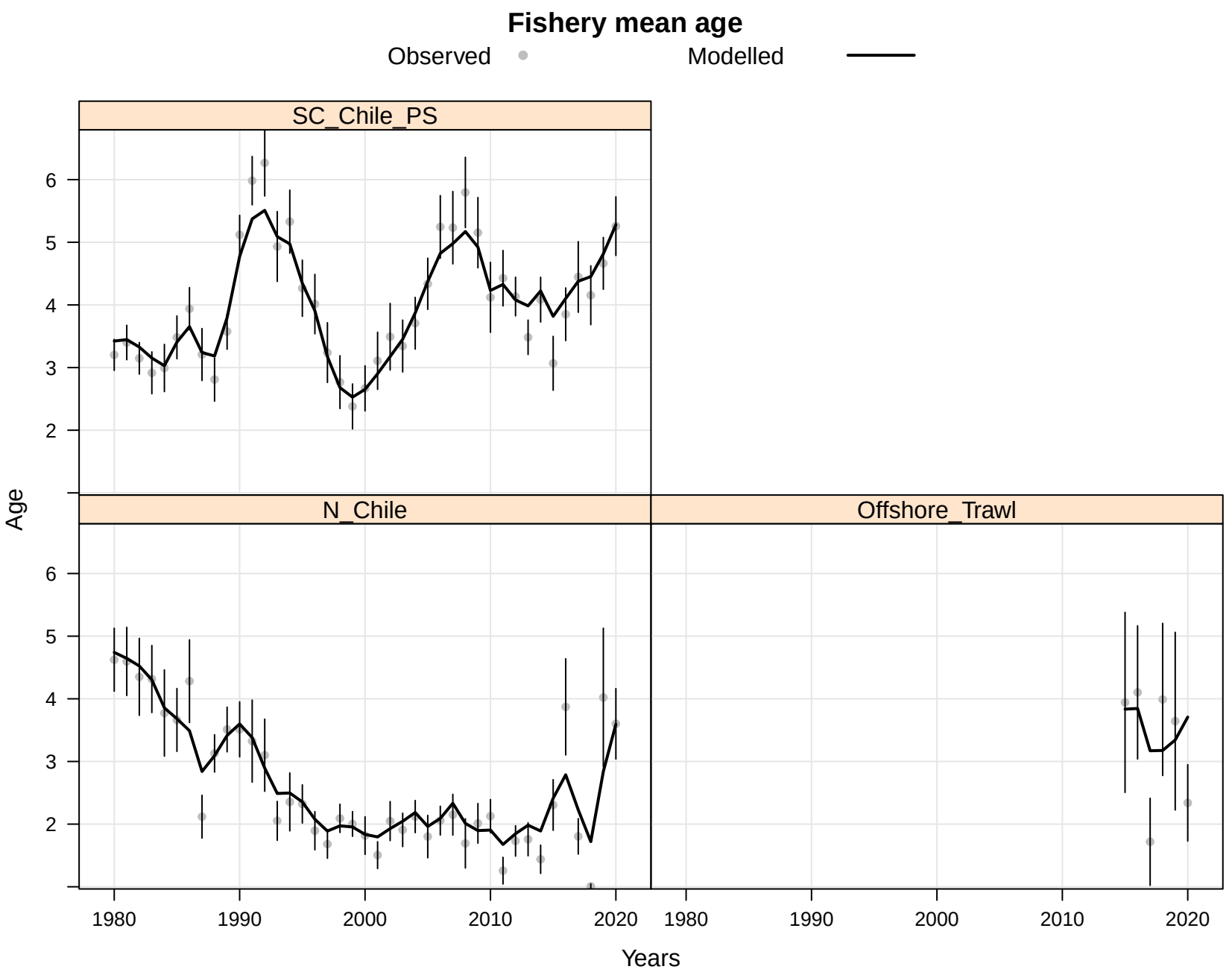

Selectivity of the survey by Pentad
Selectivity
Age
1970's
1975's
1980's
1985's
1990's
1995's
2000's
2005's
2010's
2015's
2020's
DEPM

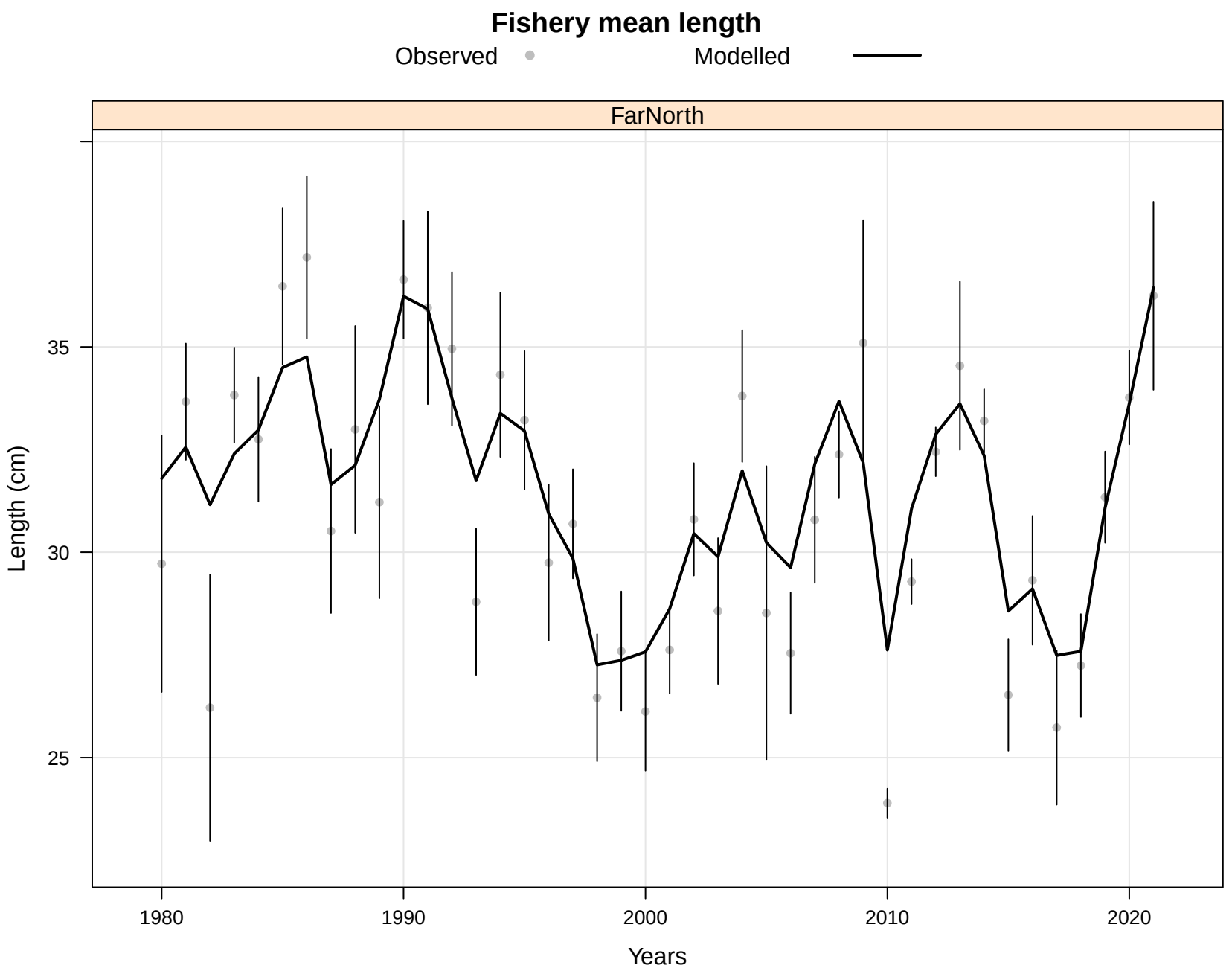

F at age
Years
Age
2021
2020
2019
2018
2017
2016
2015
2014
2013
2012
2011
2010
2009
2008
2007
2006
2005
2004
2003
2002
2001
2000
1999
1998
1997
1996
1995
1994
1993
1992
1991
1990
1989
1988
1987
1986
1985
1984
1983
1982
1981
1980
1979
1978
1977
1976
1975
1974
1973
1972
1971
1970
1 2 3 4 5 6 7 8 9 10 11 12
1.5
1.0
0.5
0.0

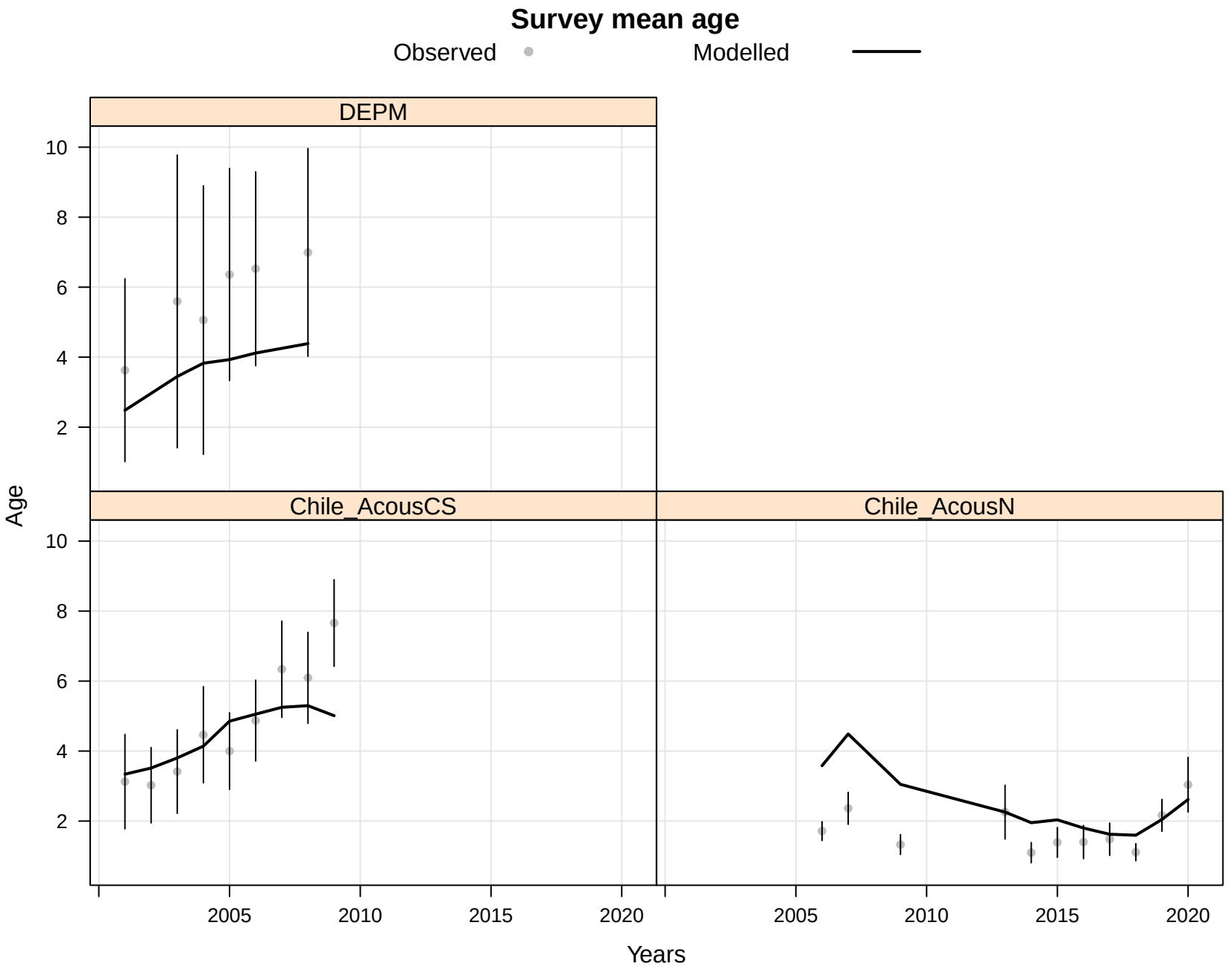
F proportion at age
Proportion of F at age
0.8
0.6
0.4
0.2
1970
1980
1990
2000
2010
2020
Years
1
2
3
4
5
6
7
8
9
10
11
12

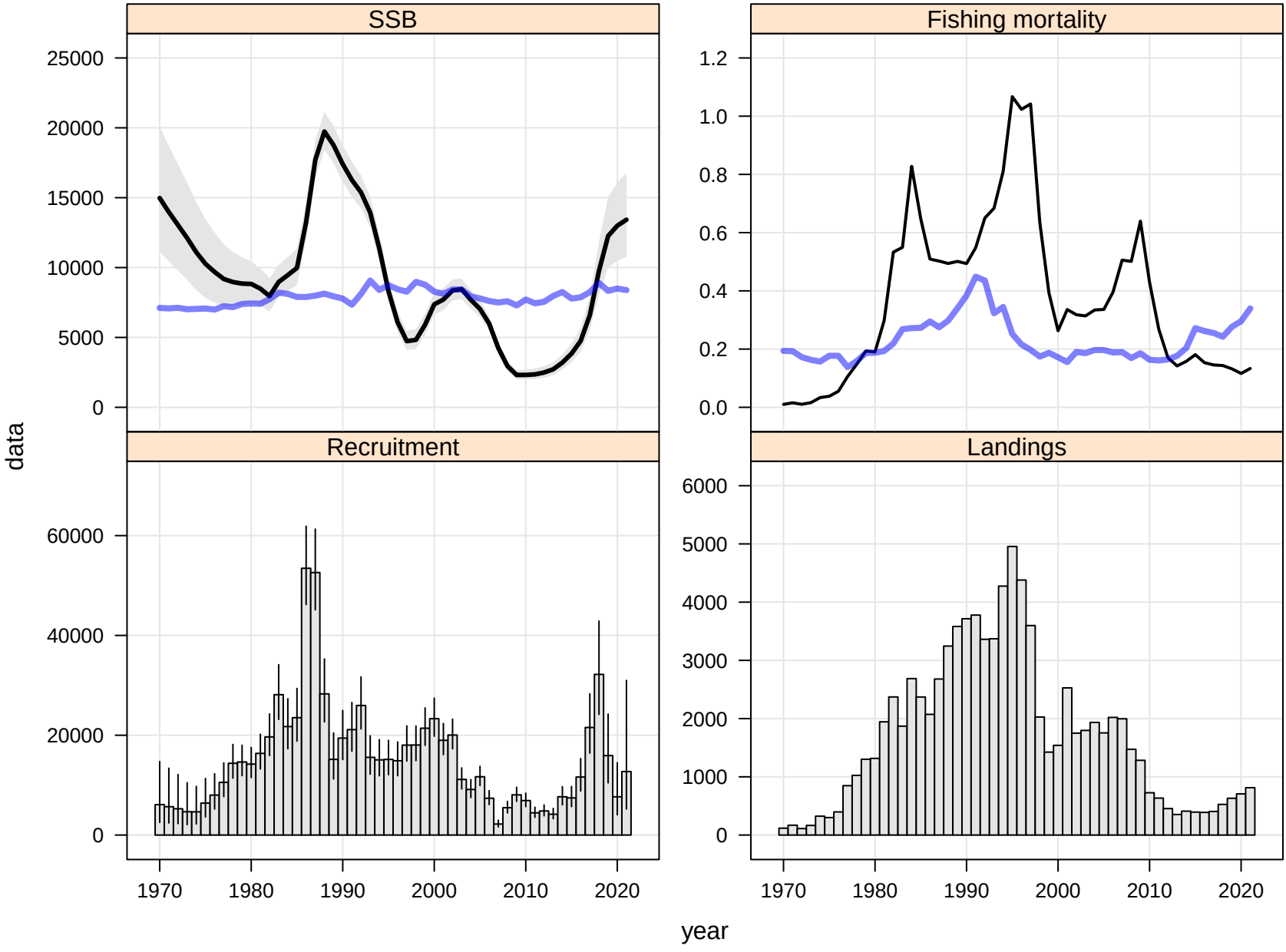

N at age
Years
50
40
30
20
10
2
4
6
8
10
12
Age
50000
40000
30000
20000
10000
0

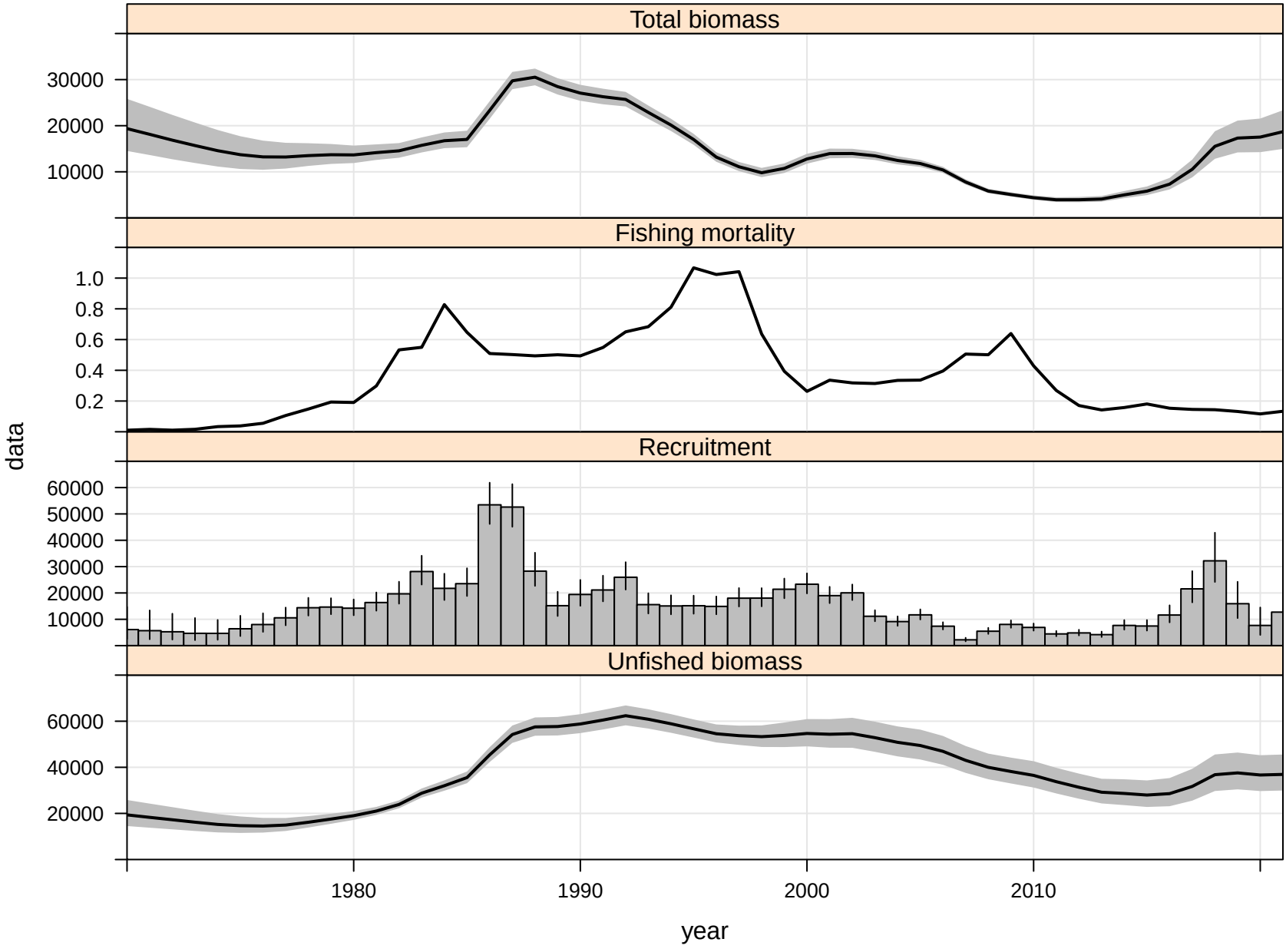

Proportion at age
Proportion of N at age
0.2
0.4
0.6
0.8
1970
1980
1990
2000
2010
2020
Years
1
2
3
4
5
6
7
8
9
10
11
12

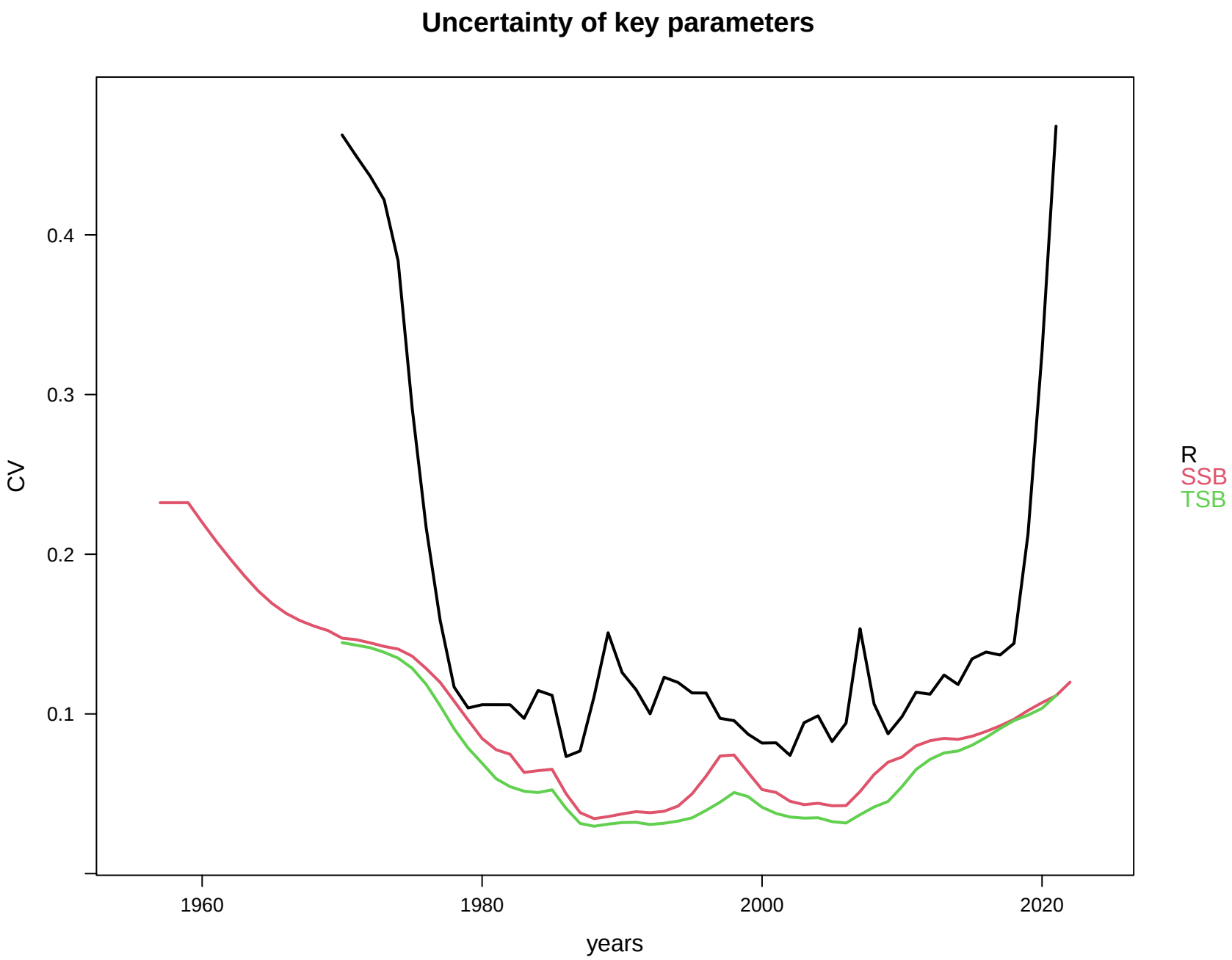

Fishery mean age
Observed
Modelled
SC_Chile_PS
N_Chile
Offshore_Trawl
Age
Years
1980
1990
2000
2010
2020
2
3
4
5
6

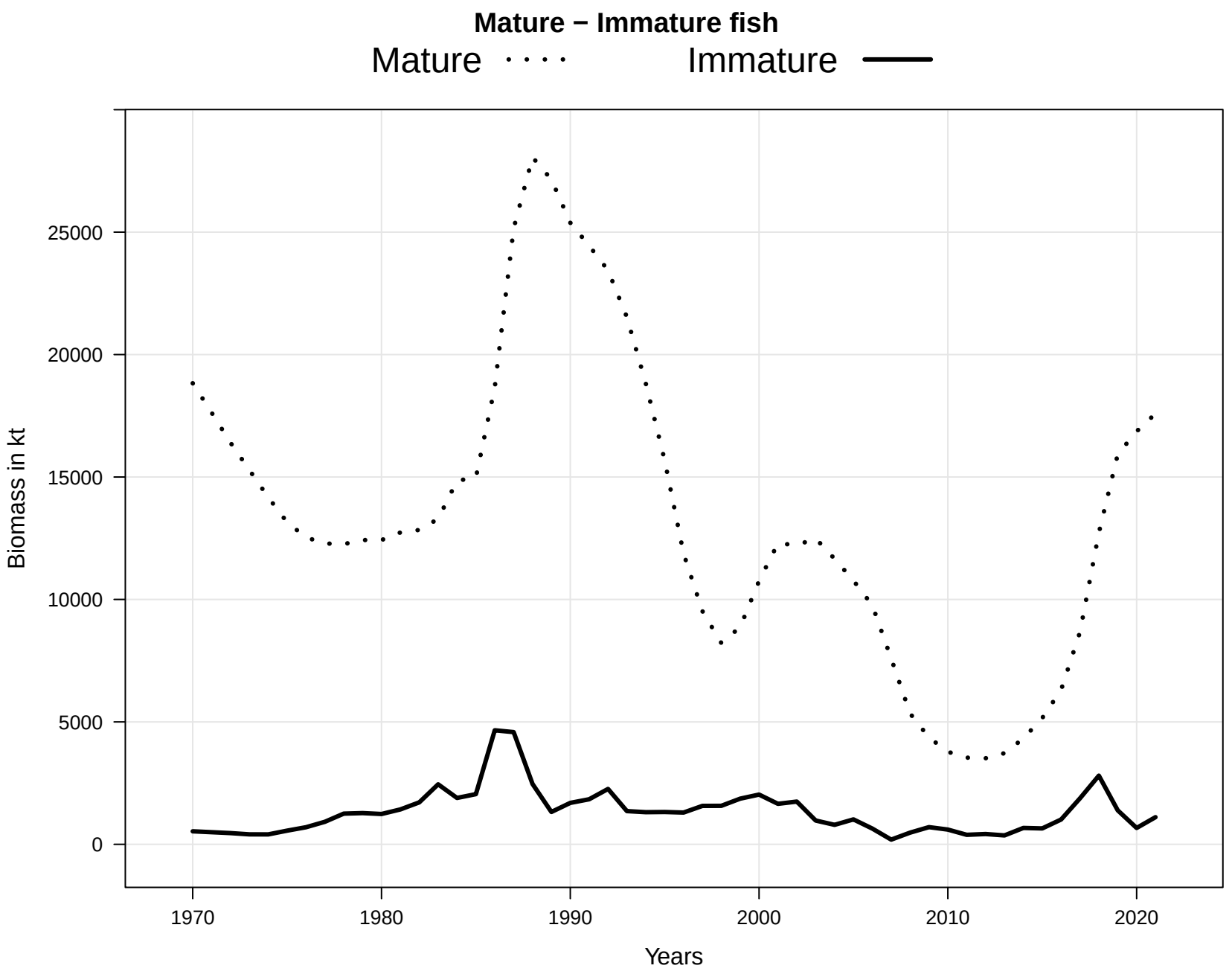

Fishery mean length
Observed
Modelled
FarNorth
Length (cm)
35
30
25
1980
1990
2000
2010
2020
Years

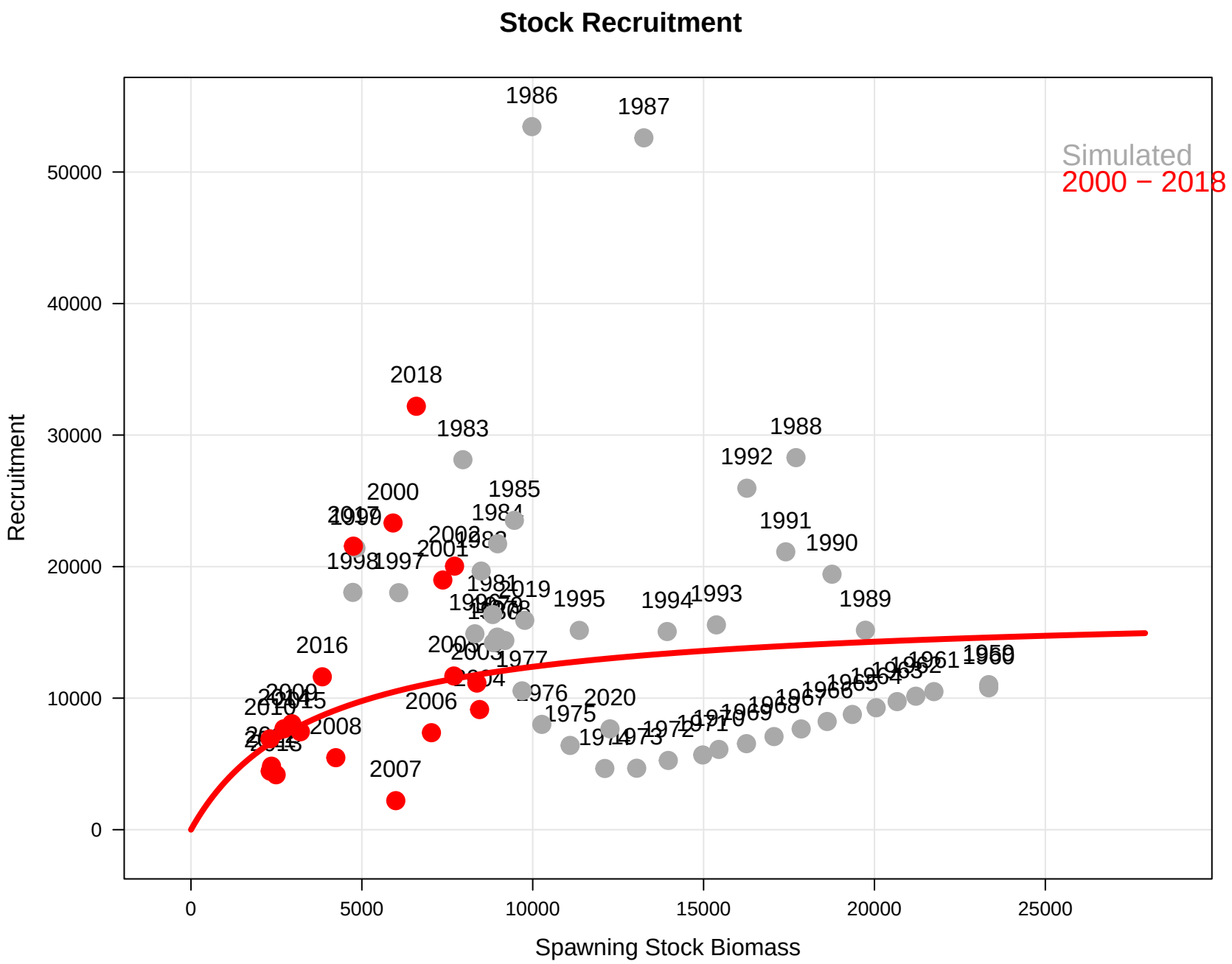

Survey mean age
Observed
Modelled
DEPM
Chile_AcousCS
Chile_AcousN
Age
Years
10
8
6
4
2
2005
2010
2015
2020

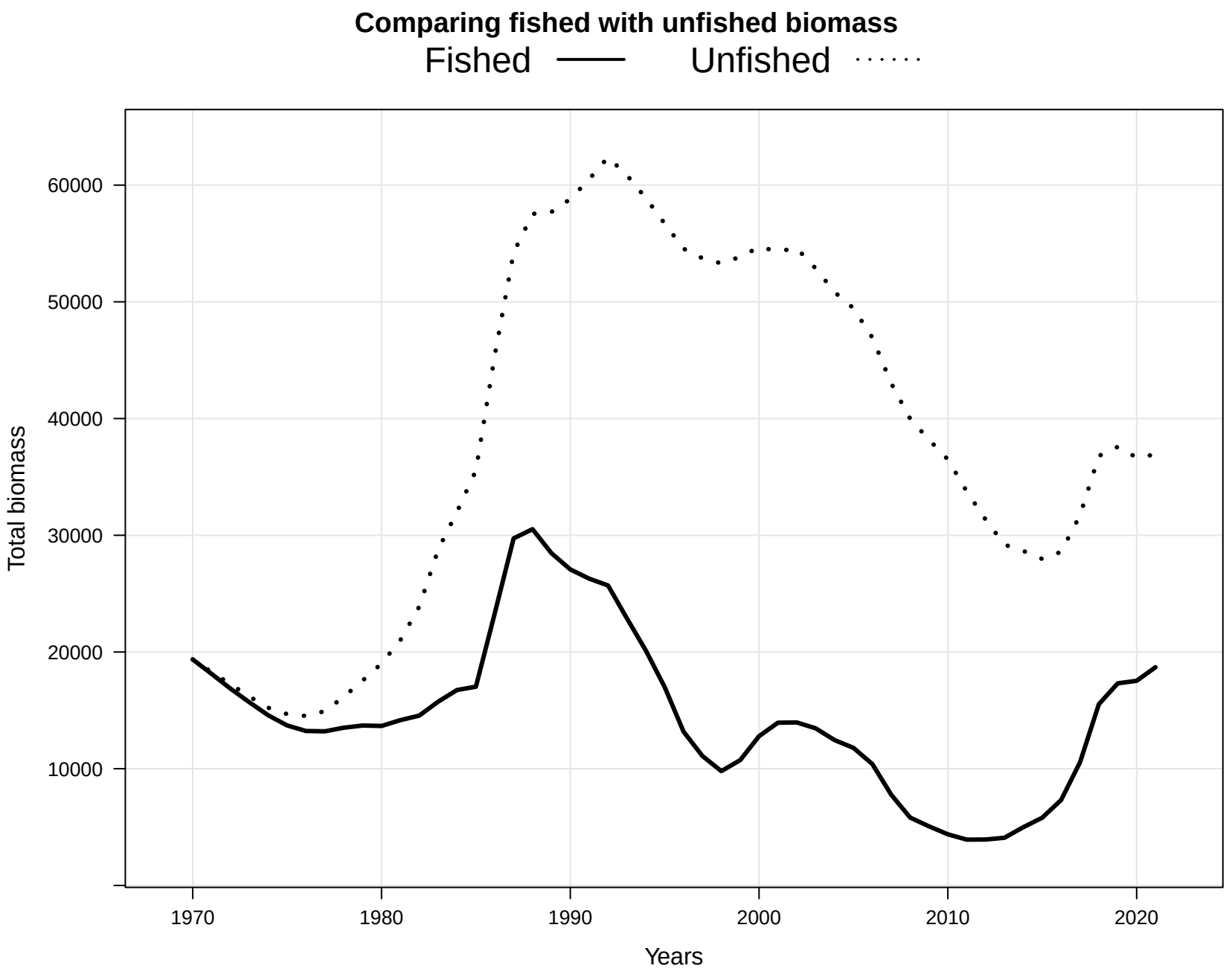

SSB
Fishing mortality
Recruitment
Landings
data
year
25000
20000
15000
10000
5000
0
1.2
1.0
0.8
0.6
0.4
0.2
0.0
60000
40000
20000
0
6000
5000
4000
3000
2000
1000
0
1970
1980
1990
2000
2010
2020

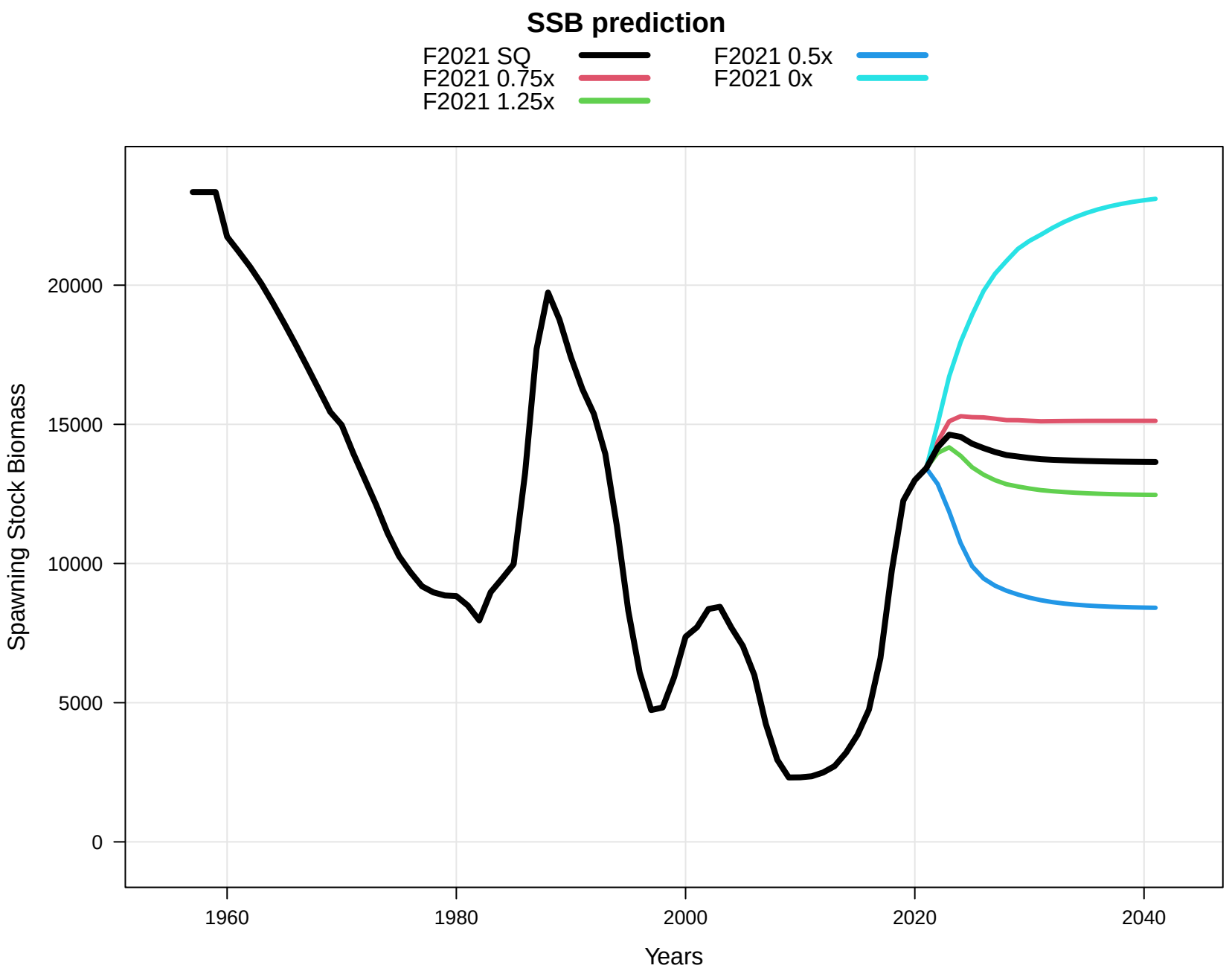

Total biomass
30000
20000
10000
Fishing mortality
1.0
0.8
0.6
0.4
0.2
Recruitment
60000
50000
40000
30000
20000
10000
Unfished biomass
60000
40000
20000
data
1980
1990
2000
2010
year

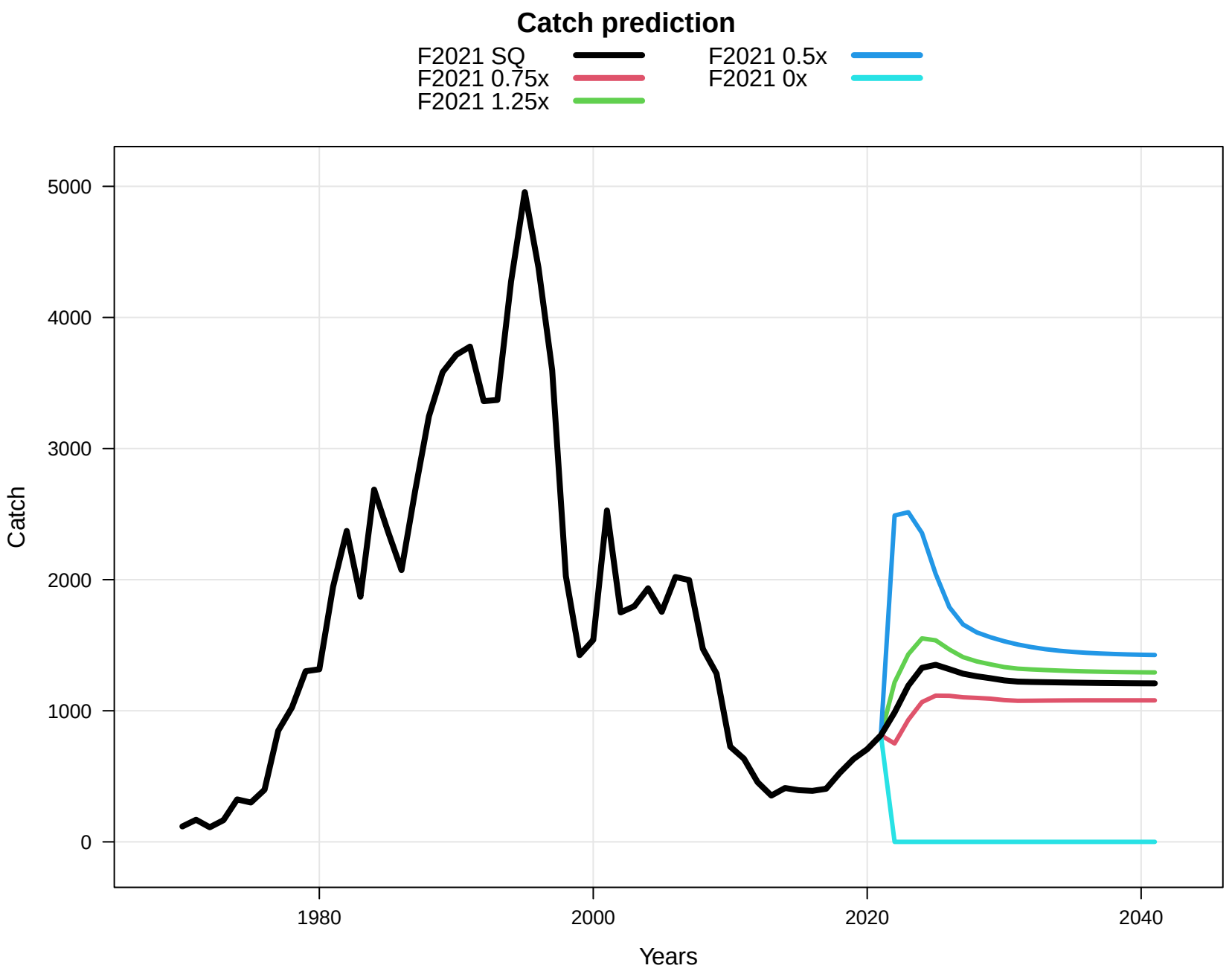

Uncertainty of key parameters
CV
0.4
0.3
0.2
0.1
1960
1980
2000
2020
years
R
SSB
TSB

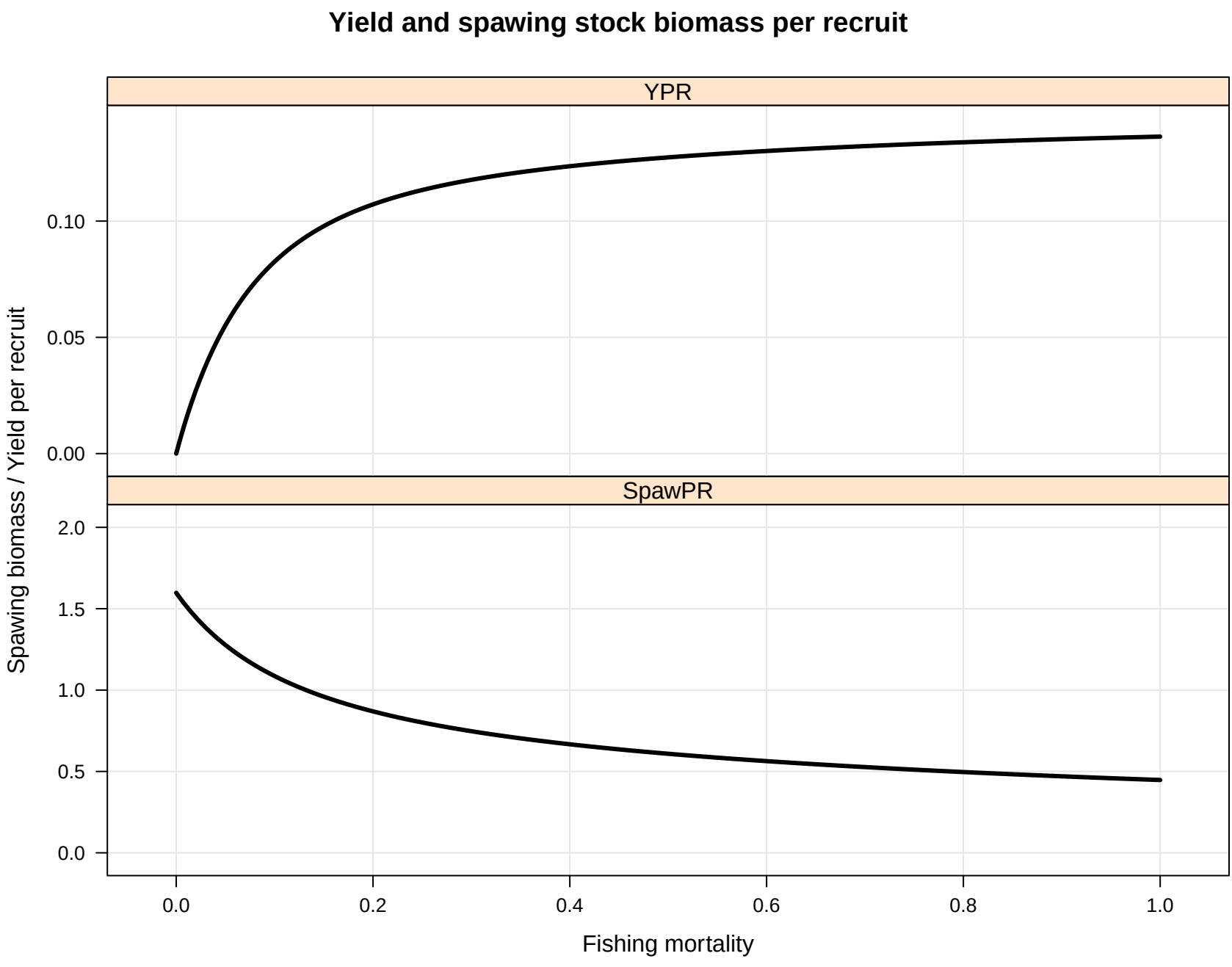

Mature − Immature fish
Mature
Immature
Biomass in kt
25000
20000
15000
10000
5000
0
1970
1980
1990
2000
2010
2020
Years

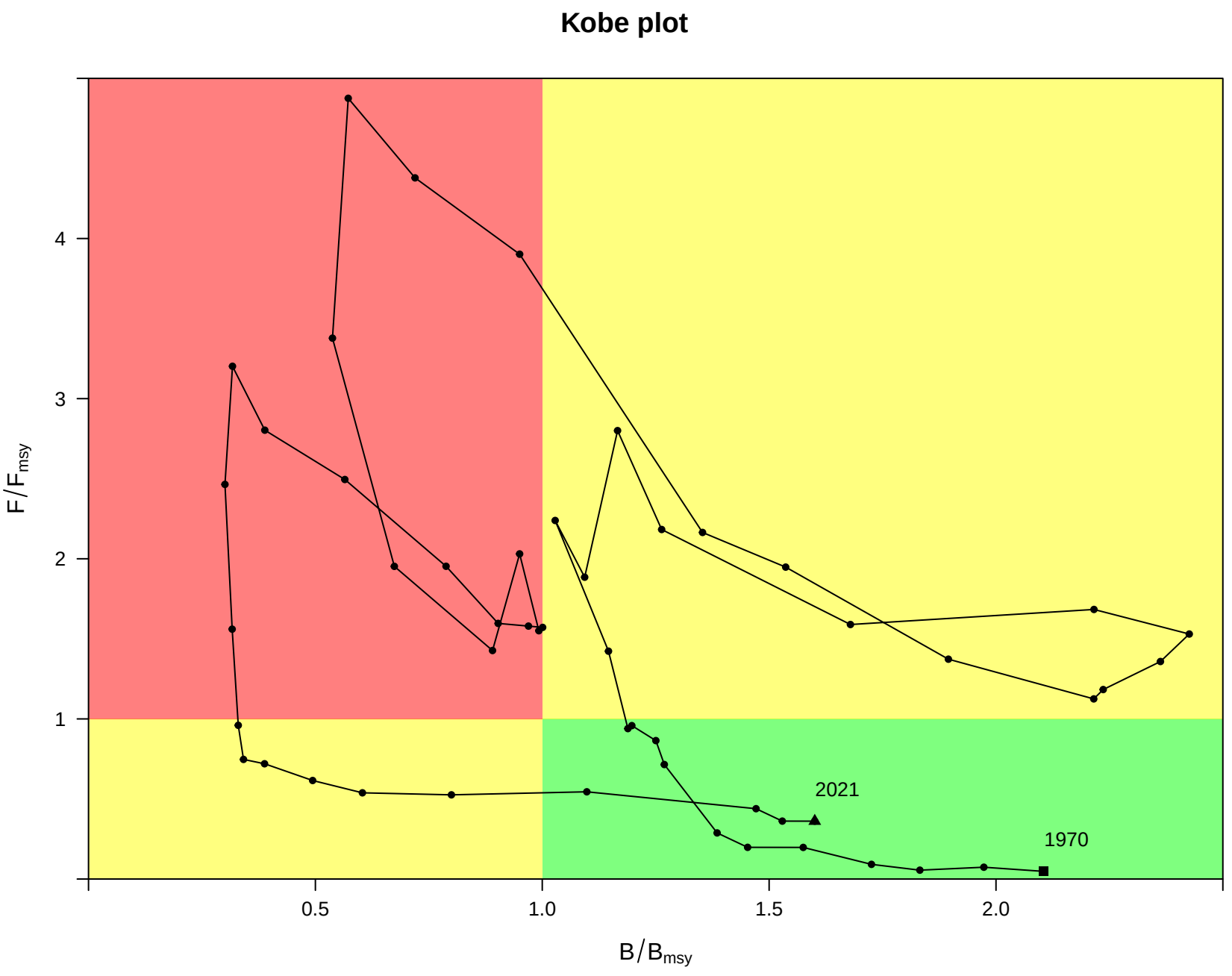

Stock Recruitment
Simulated
2000 − 2018
Recruitment
Spawning Stock Biomass
0
10000
20000
30000
40000
50000
0
5000
10000
15000
20000
25000
1986
1987
2018
1983
1988
1992
2000
1985
1991
1990
1998
1997
2001
1995
1994
1993
1989
2016
1977
2006
1976
2020
1975
2008
2007

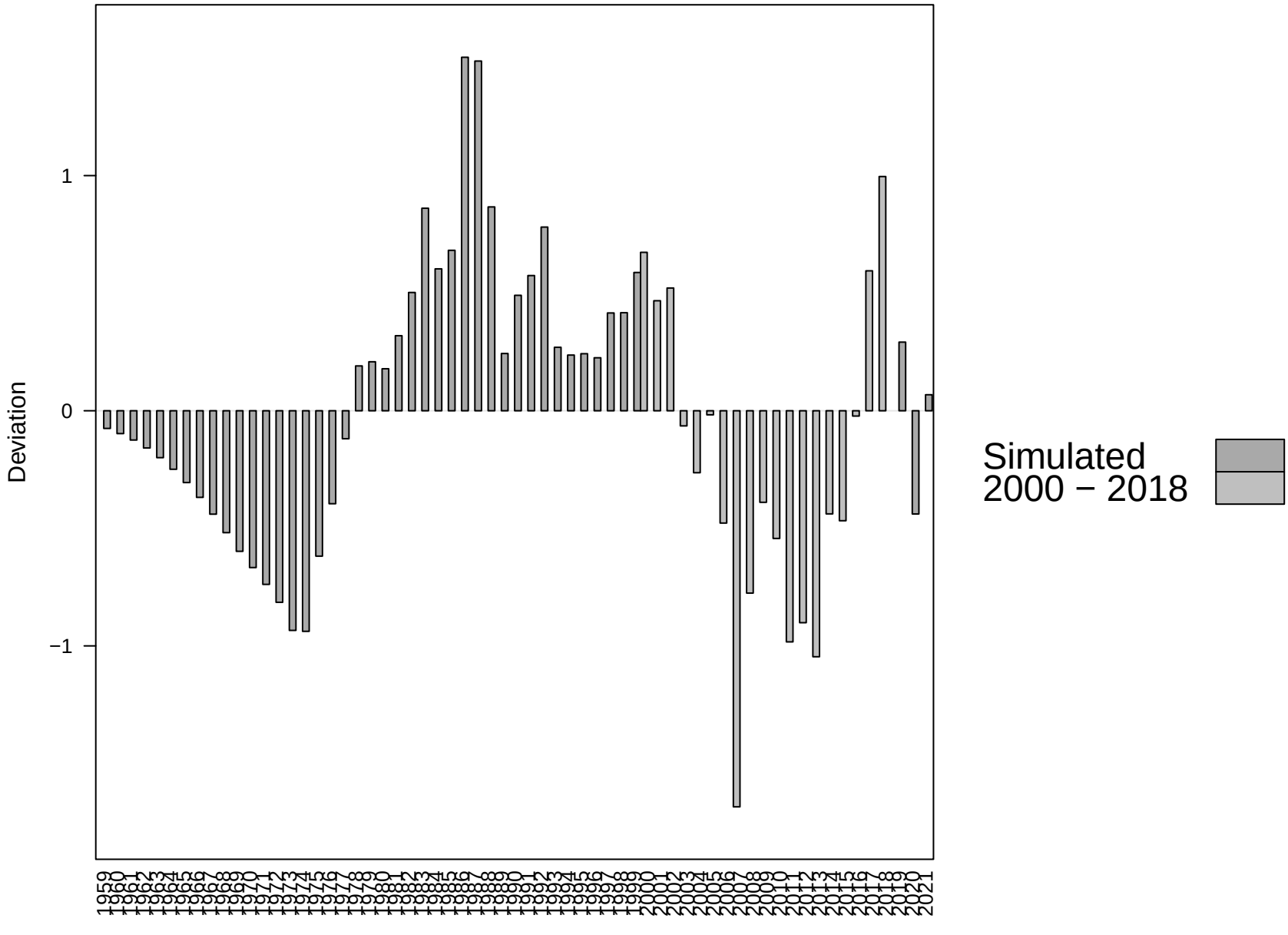

Comparing fished with unfished biomass
Fished
Unfished
Total biomass
60000
50000
40000
30000
20000
10000
1970
1980
1990
2000
2010
2020
Years

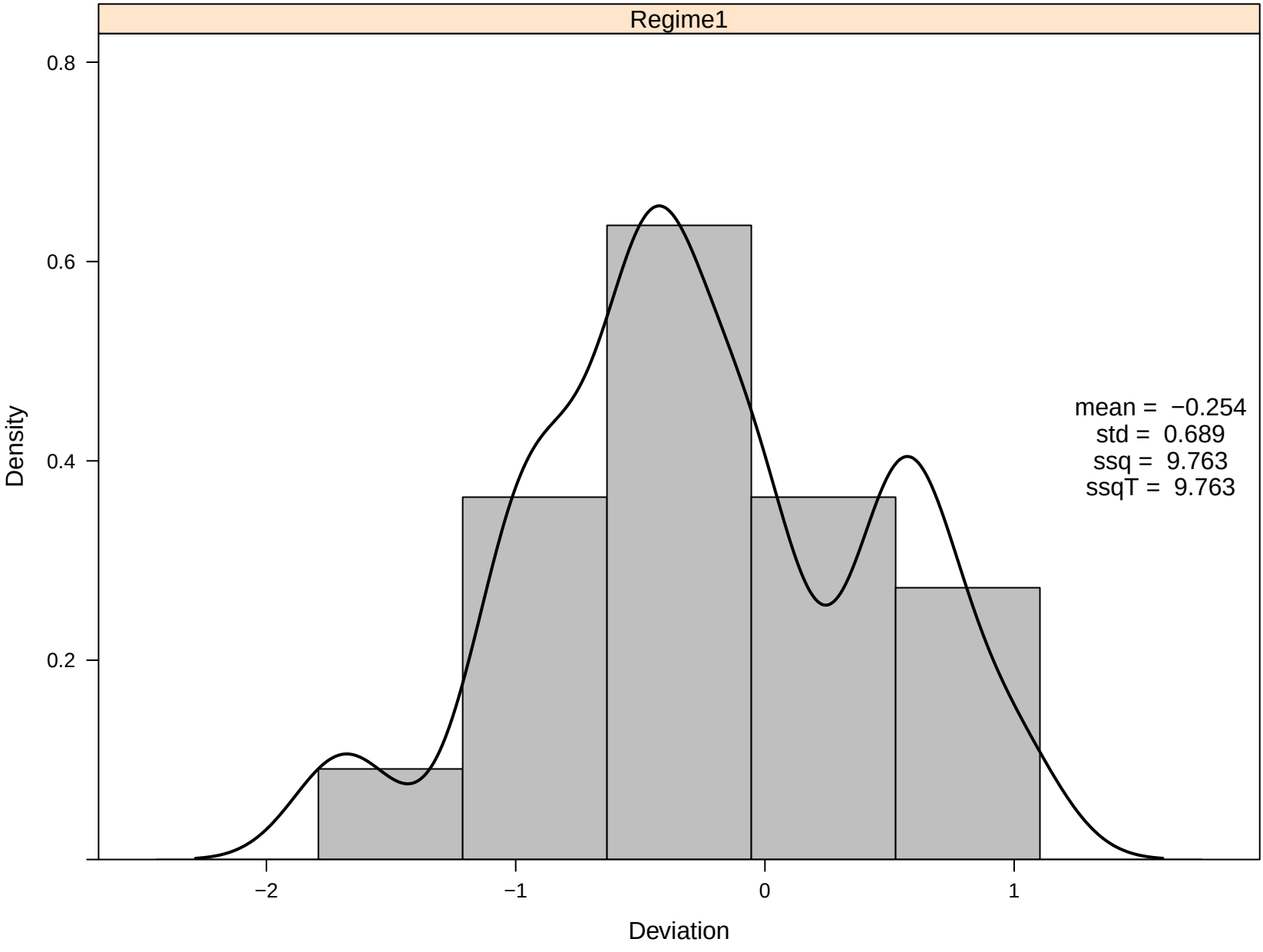

SSB prediction
F2021 SQ
F2021 0.75x
F2021 1.25x
F2021 0.5x
F2021 0x
Spawning Stock Biomass
20000
15000
10000
5000
0
1960
1980
2000
2020
2040
Years


Catch prediction
F2021 SQ
F2021 0.75x
F2021 1.25x
F2021 0.5x
F2021 0x
Catch
5000
4000
3000
2000
1000
0
1980
2000
2020
2040
Years


Yield and spawing stock biomass per recruit
YPR
SpawPR
Spawing biomass / Yield per recruit
0.00
0.05
0.10
0.0
0.5
1.0
1.5
2.0
0.0
0.2
0.4
0.6
0.8
1.0
Fishing mortality


Kobe plot
F/F_msy
B/B_msy
1
2
3
4
0.5
1.0
1.5
2.0
2021
1970


Deviation
1
0
−1
Simulated
2000 − 2018


Regime1
0.8
0.6
0.4
0.2
Density
mean = −0.254
std = 0.689
ssq = 9.763
ssqT = 9.763
−2
−1
0
1
Deviation